Biolight Patient Data Sharing Protocol

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1 Overview

The Biolight Patient Data Sharing Protocol (hereinafter referred to as the Biolight PDS Protocol) is used to export the data of the Biolight products through the TCP/IP network. As the server, the Biolight products can support multiple clients to connect to it. The Biolight PDS protocol is compatible with part of the HL7 V2.6 protocol and provides the following two types of data:

(1) Unsolicited information. The data unsolicited sent include the online and offline status of the module, the parameters supported by the module, the result values of the parameters, physiological alarm information and technical alarm information. Once the connection is established, the server periodically sends data to the client. The server that sends this message is defined as an unsolicited sending interface.

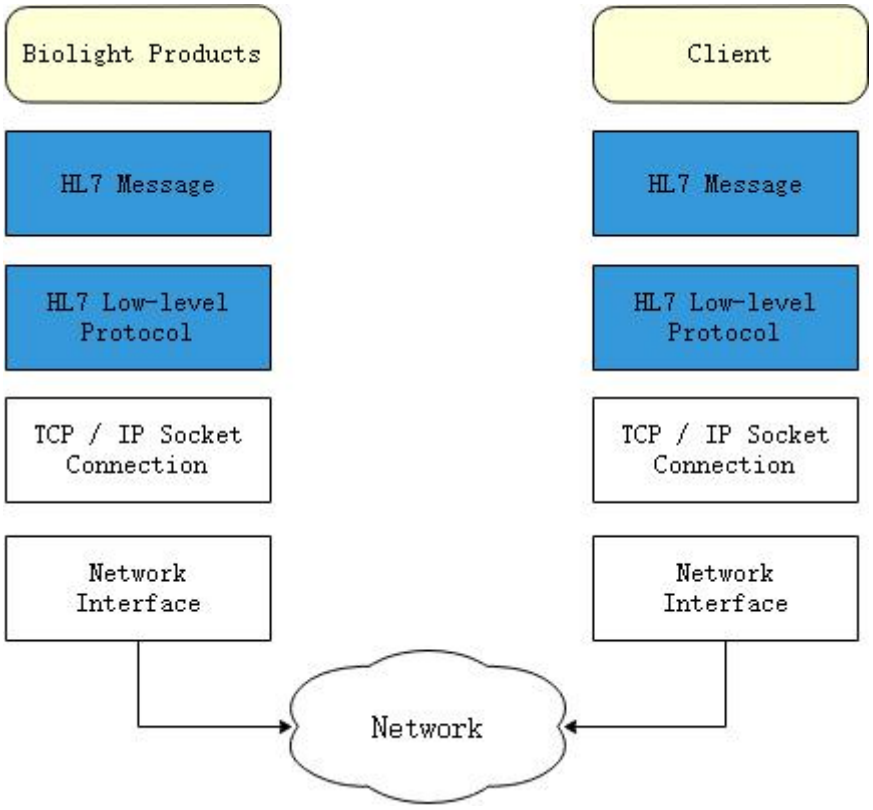
(2) Information requested by the client. Once the connection is established, the server will wait for the client to send the requested data information, and the server will send the requested specific data to the client after receiving the message.The server that sends and receives this message has two interfaces: the real-time sending interface and the query interface.

HL7 messages defined in the Paulette Patient Data Sharing Agreement include:

- ORU_R01 message: module on/off message, module support parameter message, periodic parameter message, aperiodic parameter message, physiological alarm message, technical alarm message, parameter alarm limit message, parameter alarm level message, parameter alarm switch message, waveform data message, heartbeat packet message, etc. The above messages are distinguished by different message control ids (the MSH-10 field is the message control ID. For the definition of the control ID, see *A.5 Control and Settings*).
- QRY_R02 message: data query message, which can be classified according to the query definition in the QRF_5 field. The query message types include physiological parameter data, physiological alarm data, technical alarm data, waveform data, etc. For details, see *5.2 QRY^R02 Message*.
- ORF_R04 message: indicates the data query result message. The message is the response to the QRY_R02 query message. For details, see *5.4 ORF^R04 Message*.
- QRY_A19 message: ADT patient query message. This message can be located based on the medical record number in the QRD_8 field. For details, see *7.2 ADT Query Message (QRY_A19)* .
- ADR_A19 message: ADT patient query result message. This message is the response to a QRY_A19 patient query message. For details, see 7.3 ADT Query Reply Message (ADR_A19).

2 Network communication protocol layer structure

The Biolight PDS protocol is defined above the TCP / IP layer, and the figure below shows the communication layer involved in the communication between the Biolight monitor and the client PC. The blue layer is defined in this document.



2.1 HL7 Low-Level Protocol

TCP/IP is a byte-stream protocol that does not provide message boundaries. HL7 , which belongs to application layer protocol, defines message format but does not provide any mechanism to detect message start and end. To mark the boundaries, the Minimal Lower Layer Protocol is used. On the network, a message is transmitted in the following format:

- Message start mark (1 byte) that corresponds to ASCII <VT>. Its hexadecimal value is <0x0B>.
- HL7 message of multiple bytes.

1. It includes only ISO 8859-1 characters (hexadecimal values between 0x20 and 0xFF) and <CR>. must not contain other control/non-printable characters.
2. When a Biolight product receives a message containing illegal control characters, no further processing is performed.

- Message end mark (2 bytes) that respectively corresponds to ASCII <FS> and <CR>. Their respective hexadecimal values are <0x1C> and <0x0D>.

2.2 HL7 Protocol Layer

The HL7 protocol is a communication protocol developed by the American HL7 company for data exchange between medical systems. This protocol has been widely used in many countries and regions in the world.

Biolight PDS protocol is based on HL7 protocol version 2.6. It uses only part of it.

2.3 Character Set

Biolight PDS protocol supports multiple character sets. The default character set is UTF-8. Character sets are defined in segment MSH.

MSH|^~\&|||||||ORU^R01|154155500|P|2.3.1||||||UTF-8|

The ISO8859_1 character set can be omitted if it is default.

2.4 Message Separator and Escape character

The following table lists the separators used in HL7 message.

Element	Separator
Segment	<CR> The hexadecimal value is 0X0D
Field	|
Field Repetition	~
Component	^
Sub Component	&

If the HL7 message do uses the above separator characters in field content (i.e. the patient demographics), please use the escape character "\" before them.

For example, string "A & B" should be transfer to "A\&B".

2.5 Coding System

Biolight PDS protocol uses Biolight HL7 coding system (Biolight HL7 Code, hereinafter called BHC).) Please refer Appendix A.

3. HL7 Message Type

3.1 Definition

There are many different types of HL7 messages, each with its own unique uses. Each type of HL7 message contains its own specific data content, and different data requires different types of HL7 messages to be sent. For example, parameter information and alarm information need to be sent by two different types of HL7 messages respectively.

3.1.1 The Definition Of The Message Header (MSH)

The MSH segment is the first segment of an HL7 message and defines the type of the message. The following table defines the syntax of the MSH segment.

SEQ	LEN	DT	OPT	REP	Field Name	Description
1	1	ST	R	No	Field Separator	“|”
2	4	ST	R	No	Coded Character	“^~\&”
3	227	HD	O	No	The Application Is Being Sent	
4	227	HD	O	No	Receiving Applications	
7	24	DTM	R	No	Date/Time Of The Message	
9	15	MSG	R	No	Message Type	
10	199	ST	R	No	Message Control ID	Defined In A.5 Controls And Settings
11	3	PT	R	No	Process ID	
12	60	VID	R	No	Version ID	
18	16	ID	O	Yes	Character Set	“UTF-8”

The message type is defined by the Message Type and Message Control ID fields.

3.1.2 Message Type

The value of the message type has been defined by the HL7 protocol, and the following table lists the message types used in the Paulette PDS protocol.

Message	Description
ADT	Patient Management

QRY	Inquire
ORU	Actively Sent Observation Messages

3.1.3 Message Control ID

Each message has its unique message control ID. This ID is an Arabic number defined by the Paulette PDS protocol. For details, see Appendix *A.5 Controls and Settings*.

3.2 Code System

3.2.1 OBX Segment

The OBX segment is used to send configuration information such as parameters, waveforms, alarms, alarm limits, alarm levels, etc. The following table defines the syntax of OBX segment. For details about OBX when it is used to transmit configuration information such as parameters, waveforms, alarms, alarm limits, and alarm levels, see 4.2.7 Orderly Observation OBX segment.

SEQ	LEN	DT	OPT	Field Name
2	3	ID	O	Value Type
3	705	CWE	R	Observation Mark
4	20	ST	O	Observer ID
5	99999	VARIES	O	Observation Result
6	705	CWE	O	Unit
7	60	ST	O	Reference Range
11	1	ID	R	Observed Result State
13	20	ST	O	User-Defined Access Checks
14	24	DTM	O	Date/Time Of Observation

4 Unsolicited Send Interface

This section introduces the specific content and implementation of unsolicited send interface, mainly from two aspects of communication process and data communication.

4.1 Communication Process

4.1.1 Network Connection

- **Active sending interface as server:**

As the server of TCP communication, the active sending interface always listens for connection requests from TCP clients. Once the number of clients requesting connection reaches the maximum limit, the active sending interface rejects new connection requests. After the connection is successful, the active sending interface, as a server, does not disconnect from the client. Interruption and disconnection can only be initiated by the client. In the event of an unexpected network outage, which requires the client to re-establish the connection, the data generated during the outage will also be lost. After the connection is reconnected, the active sending interface sends the data generated after the connection. In addition, it should be noted that the active sending interface is only responsible for processing the connection request, disconnection request and sending data to the client, and does not receive other requests in addition to the request, such as query requests.

If a physical network connection is interrupted during the connection, the Windows TCP SOCKET may fail to notify the active sending interface, leading us to believe that the socket is still connected properly. However, because the physical network connection is down, data cannot be sent or received properly. Therefore, if the client does not receive any data sent by the active sending interface within the set interval, the client is advised to resend the connection request. If the connection still fails to be established, take measures to rectify the network fault.

- **Active sending interface as client:**

Active Sending When the interface functions as the TCP communication client, it sends a connection request to the server every 1s until the connection is successful. After the connection is successful, the active sending interface, as the client, does not disconnect from the server. If an unexpected network interruption occurs, the active sending interface sends a connection request to the server again one second later. If the connection fails, the active sending interface tries to connect to the server once every one second until the connection is successful.

4.1.2 Network connection Settings

- Active sending interface as server: the client needs to know the IP address and port number of the server to connect. The server IP address can be viewed by clicking the [Main menu] button on the monitor and [System Info]. The port number is fixed as 6201.

- Active sending interface as client: requires us to set the IP and port number of the server to be connected. Click [Main menu] button, click [Maintenance] button, enter password 300246 or 785623 to enter the maintenance interface, click [Network Settings] button, click [HL7], [Role] select the client, and then set the server IP to be connected in [target IP], and set the server port number to be connected in [target port].

4.1.3 Sending process

After the TCP connection is established, the active sending interface sends the first ORU^R01 message to the other end through the socket, and then sends the latest monitor data (ORU^R01 message) at a set interval. The sent data and interval can be set on the HL7 setup interface, see 4.1.4 for details.

When the other end receives the data, it can send an ACK or NACK in response, but the active sending interface does not process this response message.

Aperiodic parameters such as NIBP and C.O are also sent at the set interval, and the sent value is the last manually measured value.

4.1.4 Sending content and sending interval Settings

1. Set the sending content: click [Main menu] button of the monitor, click [Maintenance] button, enter password 300246 or 785623 to enter the maintenance interface, click [Network Settings] button, click [HL7], the switch corresponding to [Send waveform] will be turned on, and the waveform data will be sent.
2. Set the sending interval: Repeat the operations described in 1 above. After entering the HL7 interface to set, click [Send parameter interval] to set a reasonable sending interval.

4.1.5 Send Content

1. After establishing the connection, the following data can be sent:
   (1) Module online data, that is, which modules are online.
   (2) Parameter configuration data, including parameter alarm switch, alarm level, alarm upper and lower limits.
   (3) Parameter data supported by the module, that is, the set of parameters supported by the online module.
   (4) Parameter data not supported by the module, that is, a collection of parameters not supported by the online module.
   (5) Physiological parameter data of periodic modules (sent once after establishing a connection, and then periodically sent according to the set sending interval).
2. The data to be sent periodically according to the set sending interval is as follows:
   (1) Physiological parameter data of periodic modules.
   (2) Module configuration and device status: includes device standby status, maximum physiological and technical alarm levels, device alarm status, ECG lead type, PR source, RR source.

3. The following data will be sent when triggered:
   (1) Non-periodic module physiological parameter data, such as: NIBP, C.O. data when the measurement is completed.
   (2) Module online information, one of the modules is online, send once.
   (3) Module offline information, one of the modules offline, send once.
   (4) Parameter information supported by the module and parameter information not supported by the module (generally sent with the module uplink and down information).
   (5) When the parameter configuration data (parameter alarm switch, alarm level, alarm upper and lower limits) changes, it will be sent once.
4. The parameter waveform data is sent periodically, with a transmission period of 1 second.
5. Physiological alarm information and technical alarm information are sent periodically, with a transmission period of 1 second.

4.2 ORU_R01 Message

The ORU in ORU_R01 is the unsolicited observation message, R01 is the event to be triggered, and the type of the message sent by the unsolicited send interface is the ORU unsolicited observation message. An HL7 message contains data for one patient.

The ORU_R01 message is divided into five parts: message header, software segment, user authentication, patient result, and continuous pointer.

The patient results are divided into two parts: the patient and the orderly observation.The patient part mainly includes patient identification (PID), additional demographic information (PD1), description and annotation (NTE), observations (OBX), visit (patient visit (PV1) and visit additional information (PV2)); Order observation mainly includes general order (ORC), observation report ID (OBR), description and annotation (NTE), observation role (ROL), timing / quantity (timing / quantity (TQ1), order sequence (TQ2)), contact data (CTD), observation (observation / result (OBX), description and annotation (NTE)), financial affairs (FT1), clinical trial identification (CT1), sample (sample (SPM), sample involved result (OBX)).

The sections used this time are:

- MSH _ 1 (field separator), MSH_2 (code character), MSH_9 (message type), MSH_10 (message control ID), MSH_11 (process ID), MSH_12 (version ID) for the message header (MSH);
- PID_3 (list of patient identifiers), PID_5 (patient name), PID_7 (date/time of birth), PID_8 (sex) of patient identification(PID);
- PV1_2 (patient class), PV1_3 (designated patient location), and PV1_7 (receiving doctor) of patient treatment (PV1);
- OBR_4 (set ID-OBR), OBR_7 (observation date / time) of observation ID (OBR);
- OBX_2 (valuetype), OBX_3 (observation identifier), OBX_4 (observed sub-ID), OBX_5 (observation results), OBX_11 (observation results status) of the observation results (OBX);

4.2.1 Format

An HL7 message in the unsolicited send interface contains data from a bed or a patient.

Patient_1	MSH Segment	PID Segment	PV1 Segment	OBR Segment	OBX Segment	OBX Segment	...	OBX Segment

Patient_2	MSH Segment	PID Segment	PV1 Segment	OBR Segment	OBX Segment	OBX Segment	...	OBX Segment

.....

Patient_N	MSH Segment	PID Segment	PV1 Segment	OBR Segment	OBX Segment	OBX Segment	...	OBX Segment

4.2.2 MSH Segment

MSH segment is first segment of HL7 message, it defines the message type.

SEQ	LEN	DT	OPT	REP	Field Name	Description
1	1	ST	R	No	Field Separator	|
2	4	ST	R	No	Encoding Characters	^~\&
9	15	MSG	R	No	Message Type	ORU^R01^ORU_R01
10	199	ST	R	No	Message Control ID	An identifier that contains a number or otherwise uniquely identifies a message, for example:MSH|^~\&|||||||ORU^R01^ORU_R01|1004|P|2.6|||||UTF-8 1004 represents Aperiodic parameter. For details, see *Appendix A.5*
11	3	PT	R	No	Processing ID	P
12	60	VID	R	No	Version ID	2.6

4.2.3 PID Segment

SEQ	LEN	DT	OPT	REP	Field Name	Description
3	250	CX	R	Yes	Patient Identifier List	Patient Medical Record Number
5	250	XPN	R	Yes	Patient Name	<First Name>^<Last Name>
7	24	DTM	O	No	Date/Time of Birth	YYYYMMDD
8	1	IS	O	No	Sex	M:Male F:Female

						U:Unknown

4.2.4 Part of the patient's OBX Segment

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	NM
3	705	CWE	R	No	Observation Identifier	Patient record number Example： OBX||NM|4201^Height^BHC||150||||||F The field 4201^Height^BHC indicates that the message is sending patient height data, see *Appendix A.4* for details
5	99999	VARIES	O	Yes	Observation Results	Patient information value Example： OBX||NM|4201^Height^BHC||150||||||F The field 150 indicates that the patient's height is 150cm
6	705	CWE	O	No	Units	This field is the unit of the current Observation Results, represented as: Unit ID ^ Unit ^ BHC. The unit ID is detailed in *A.9 Units ID*.
11	1	ID	R	No	Observation Results Status	F：Stand for “Final”

4.2.5 PV1 Segment

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	1	IS	R	No	Patient Class	I:Inpatient O:Outpatient
3	80	PL	O	No	Assigned Patient Location	<point of care> ^<room>^<bed> <bed> divided into: <OfficeName>&<bed id>&<IP>&<IPSeq>&0 <IP>:the bedside device IP Address; <IPSeq>:the telemetry transmitter’s sequence number.
7	250	XCN	O	Yes	Attending Doctor	Doctor ID
18	2	IS	O	No	Patient Type	“N”: Neonate “A”: Adult “P”: Pediatric

						“U”: Unknow

4.2.6 OBR Segment

SEQ	LEN	DT	OPT	REP	Field Name	Description
4	705	CWE	R	No	Universal Service ID	Monitor
7	24	DTM	O	No	Observation Date/Time	yyyymmddhhmmss Example:OBR||||Monitor|||20210914161435 The observation time is:16:14:35,14 September 2021

4.2.7 Orderly Observation OBX segment

4.2.7.1 Online Modules

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	CE
3	705	CWE	R	No	Observation Identifier	Parameter attribute identification ID (this message is an online module) Example:OBX||CE|5^^BHC||5001^ECG^BHC||||||F The5^^BHC represents a module for this message, see *Appendix A.2*
5	99999	VARIES	O	Yes	Observation Results	The online module identification ID indicates which modules are online Example:OBX||CE|5^^BHC||5001^ECG^BHC||||||F The 5001 ^ ECG ^ BHC represents ECG for details in *Appendix A.3*.
11	1	ID	R	No	Observation Results Status	F

The OBX segment indicates that the ECG module is online. The format is as follows:

Format: Parameter attribute ID^ ^BHC
Parameter attribute ID: Defined in Appendix A.2
5: Module on-line

OBX||CE|5^^BHC||5001^ECG^BHC|||||F

Format: Module ID^Module name^BHC
Module ID: Defined in Appendix A.3
5001: Indicates that the ECG module is online

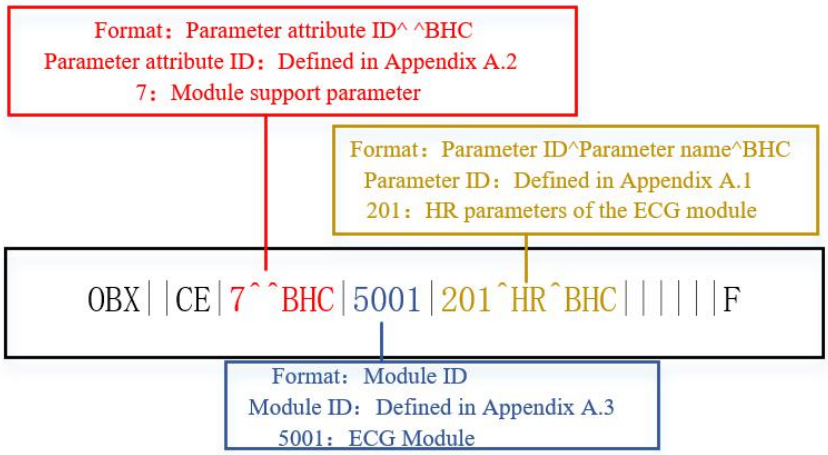
4.2.7.2 Supported Parameters

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	CE
3	705	CWE	R	No	Observation Identifier	Parameter attribute identification ID(this message is a supported parameter) OBX||CE|7^^BHC|5001|201^HR^BHC||||||F The 7^^BHC represents the supported parameters sent by this message. For details, please refer to *Appendix A.2*.
4	20	ST	O	No	Observation Sub-ID	Module identification ID (specifically which module) Example: OBX||CE|7^^BHC|5001|201^HR^BHC||||||F The 5001 represents the ECG module, That is, the message is a parameter supported by the ECG module
5	99999	VARIES	O	Yes	Observation Results	Specific parameter identification ID supported by the module， Example: OBX||CE|7^^BHC|5001|201^HR^BHC||||||F The 201^HR^BHC represents HR parameters for details in *Appendix A.1*.
11	1	ID	R	No	Observation Results Status	F

This message is included in a module online message, typically an online module message (OBX message) followed by information about all parameters supported by the online module

(OBX message).

The OBX segment indicates that the ECG module supports HR parameters in the following format:

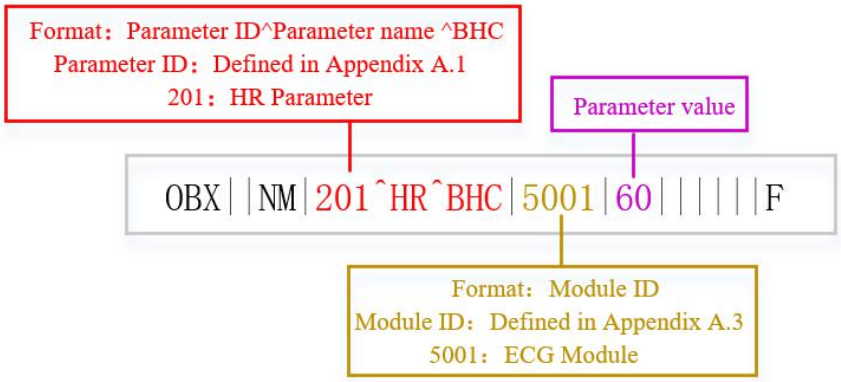


4.2.7.3 Periodic Parameter Values

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	NM
3	705	CWE	R	No	Observation Identifier	Parameter identification ID， Example: OBX||NM|201^HR^BHC|5001|60||||||F The 201^HR^BHC represents HR parameters for details in *Appendix A.1*.
4	20	ST	O	No	Observation Sub-ID	Module identification ID (specifically which module), Example: OBX||NM|201^HR^BHC|5001|60||||||F The 5001 represents the ECG module, please refer to *Appendix A.3* for details
5	99999	VARIES	O	Yes	Observation Results	Parameter values ， Example:OBX||NM|201^HR^BHC|5001|60|||||F The 60 is the value of the HR parameter of the ECG module at this time.
6	705	CWE	O	No	Units	This field is the unit of the current Observation Result. If this field is blank, it means that the observation result unit is the default unit in *A.1 parameter ID*; On the contrary, the units of the observation results are represented as: unit ID ^ unit ^ BHC,

						and the unit ID is defined in *A.9 Units ID*.
11	1	ID	R	No	Observation Results Status	F

The OBX segment indicates that the HR parameter value of the ECG module is 60 bpm in the following format:



4.2.7.4 Physiological Alarm

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	CE
3	705	CWE	R	No	Observation Identifier	Alarm level (consistent with the set alarm configuration)
5	20	ST	O	No	Observation Results	Parameter alarm identification OBX||CE|3||10151^SpO2 High^BHC||||||F|||2022051210391, 10151^SpO2 High^BHC in this field refers to the physiological alarm content of SPO2. For the specific alarm content, please refer to the alarm information corresponding to 10151 in Appendix A.6
11	99999	VARIES	O	Yes	Observation Results Status	F
14	1	ID	R	No	Date/Time of the Observation	yyyymmddhhmmss

The OBX segment represents an advanced alarm with SpO2 too high at time 2022051210391, expressed in the following format:

Alarm Level ID
0:Tips
1:Low level alarm
2:Middle alarm
3:High alarm

Alarm occurrence time

OBX||CE|3||10151^SpO2 High^BHC|||||F|||2022051210391

Format：Physiological alarm ID^Alarm content^BHC
Physiological alarm ID：Defined in Appendix A.6
10151：The SpO2 is too high

4.2.7.5 Technical Alarm

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	CE
3	705	CWE	R	No	Observation Identifier	Alarm level
5	20	ST	O	No	Observation Results	Parameter alarm identification (same as physiological alarm) See Appendix A.7 for details
11	99999	VARIES	O	Yes	Observation Results Status	F
14	1	ID	R	No	Date/Time of the Observation	Yyyymmddhhmmss

The OBX segment represents a technical alarm for ECG lead loss at time 2022051210391, expressed in the following format:

Alarm Level ID
0:Tips
1:Low level alarm
2:Middle alarm
3:High alarm

Alarm occurrence time

OBX||CE|2||20016^ECG Lead Off^BHC|||||F|||2022051210391

Format：Technical alarm ID^Alarm content^BHC
Technical alarm ID：Defined in Appendix A.7
20016：Indicates that the ECG Lead Off

4.2.7.6 System Parameter

SEQ	LEN	DT	OPT	REP	Field Name	Description

2	3	ID	O	No	Value Type	Guardian name:ST Other:CE
3	705	CWE	R	No	Observation Identifier	Specific system parameters Monitor name Standby state Highest level of physiological alarm Highest level of technical alarm System alarm status ECG lead type PR source RR source
5	20	ST	O	No	Observation Results	Monitor name:Monitor Standby state: 0（Non-standby） 1（Standby） Physiological (technical) highest alarm level: 0（Unknown）; 1（Prompt）; 2（Low）; 3（Middle）; 4（High） System alarm status: 0（Normal）; 1（Pause ）; 2（Reset）; 3（Close）; 4（Default） ECG lead type: 0（Unknown）; 1（3 leads）; 2（5 leads）; 3（6 leads）; 4（12 leads） PR source: 0（Unknown）; 1（SPO2）; 2（SPO2L）; 3（IBP）; RR source: 0（Unknown）; 1（ECG）; 2（CO2）;

						3（RM）; 4（SPO2）
11	99999	VARIES	O	Yes	Observation Results Status	F

The following OBX segments indicate that the ECG lead type is 3 leads, in the following format:

Format：Parameter ID^Parameter name^BHC
Parameter ID：Defined in Appendix A.1
4024：ECG Lead Type

OBX||CE|4024^ECG Lead Type^BHC||1^3 Lead^BHC|||||F

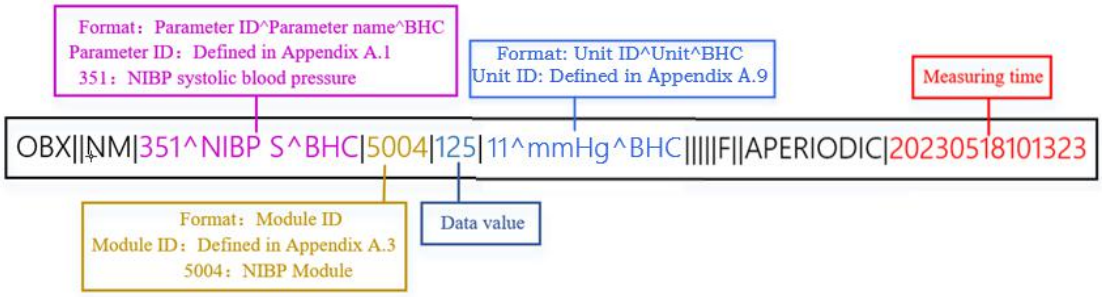
Format：Lead type ID^Lead type name ^BHC
Lead type ID：Defined in Appendix A.1
1：3 Lead

4.2.7.7 Aperiodic Parameter Values

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	NM
3	705	CWE	R	No	Observation Identifier	Specific parameter identification
4	20	ST	O	No	Observation Sub-ID	Module identification (specifically which module)
5	99999	VARIES	O	Yes	Observation Results	Parameter values
6	705	CWE	O	No	Units	This field is the unit of the current Observation Result. If this field is blank, it means that the observation result unit is the default unit in *A.1 parameter ID*; On the contrary, the units of the observation results are represented as: unit ID ^ unit ^ BHC, and the unit ID is defined in *A.9 Units ID*.
11	1	ID	R	No	Observation Results Status	F
13	20	ST	O	No	User-defined access query	APERIODIC
14	24	DTM	O	No	Date/Time of	yyyymmddhhmmss

					the Observation	Same time as OBR_7 field

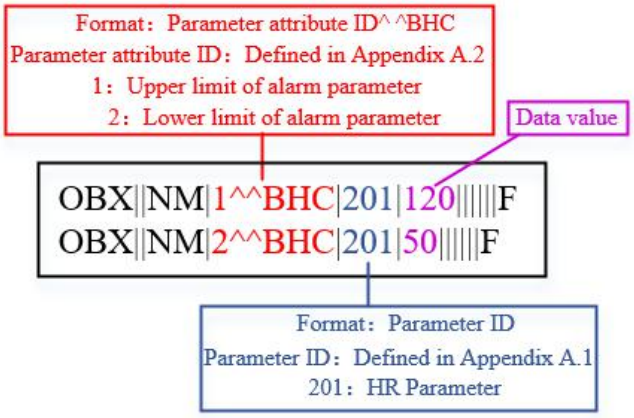
The following OBX segment represents a diastolic pressure of 125 mmHg for the aperiodic parameter NIBP measured at 20230518101323, expressed in the following format:



4.2.7.8 Parameter alarm limit

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	NM
3	705	CWE	R	No	Observation Identifier	Upper limit of parameter:1^^BHC Lower limit of parameter:2^^BHC
4	20	ST	O	No	Observation Sub-ID	Parameter identification ID
5	99999	VARIES	O	Yes	Observation Results	Alarm upper or lower limit value
6	705	CWE	O	No	Units	This field is the unit of the current Observation Result. If this field is blank, it means that the observation result unit is the default unit in *A.1 parameter ID*; On the contrary, the units of the observation results are represented as: unit ID ^ unit ^ BHC, and the unit ID is defined in *A.9 Units ID*.
11	1	ID	R	No	Observation Results Status	F

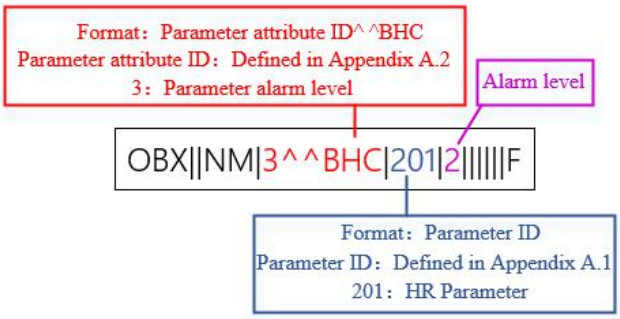
The following OBX segment represents the HR heart rate parameter alarm threshold of [50,120] bpm, expressed in the following format:



4.2.7.9 Parameter alarm level

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	3	ID	O	No	Value Type	NM
3	705	CWE	R	No	Observation Identifier	Alarm level identifier:3^^BHC
4	20	ST	O	No	Observation Sub-ID	Parameter identification ID
5	99999	VARIES	O	Yes	Observation Results	Alarm level value: 0:Tips 1:Low level alarm 2:Middle alarm 3:High alarm
11	1	ID	R	No	Observation Results Status	F

The following OBX segment indicates that the alarm level of HR heart rate parameter is intermediate alarm, and the format is as follows:

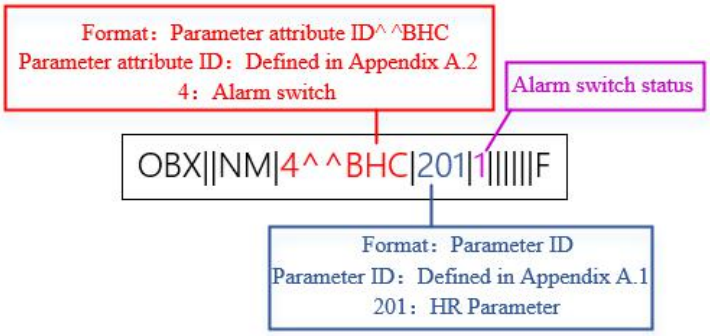


4.2.7.10 Parameter alarm switch

SEQ	LEN	DT	OPT	REP	Field Name	Description

2	3	ID	O	No	Value Type	NM
3	705	CWE	R	No	Observation Identifier	Alarm switch identifier:4^^BHC
4	20	ST	O	No	Observation Sub-ID	Parameter identification ID
5	99999	VARIES	O	Yes	Observation Results	Alarm switch value: 1（Switch on） 0（Switch off）
11	1	ID	R	No	Observation Results Status	F

The following OBX segment indicates that the alarm switch of HR heart rate parameter is on, and the format is as follows:



4.2.8 Example

1. Within 1 second after establishing the connection, send the online module parameter flag first.

```
MSH|^~\&|||||20230518094209||ORU^R01^ORU_R01|1001|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518094209
OBX||CE|5^^BHC||5001^ECG^BHC||||||F
OBX||CE|7^^BHC|5001|201^HR^BHC||||||F
OBX||CE|7^^BHC|5001|219^PVCs^BHC||||||F
OBX||CE|7^^BHC|5001|202^QTC^BHC||||||F
OBX||CE|7^^BHC|5001|203^DELTAQTC^BHC||||||F
OBX||CE|7^^BHC|5001|204^ST-I^BHC||||||F
OBX||CE|7^^BHC|5001|205^ST-II^BHC||||||F
OBX||CE|7^^BHC|5001|206^ST-III^BHC||||||F
OBX||CE|7^^BHC|5001|207^ST-aVR^BHC||||||F
```

OBX||CE|7^^BHC|5001|208ST-aVL^BHC||||||F
OBX||CE|7^^BHC|5001|209ST-aVF^BHC||||||F
OBX||CE|7^^BHC|5001|210ST-V1^BHC||||||F
OBX||CE|7^^BHC|5001|211ST-V2^BHC||||||F
OBX||CE|7^^BHC|5001|212ST-V3^BHC||||||F
OBX||CE|7^^BHC|5001|213ST-V4^BHC||||||F
OBX||CE|7^^BHC|5001|214ST-V5^BHC||||||F
OBX||CE|7^^BHC|5001|215ST-V6^BHC||||||F
OBX||CE|5^^BHC||5002^SPO2^BHC||||||F
OBX||CE|7^^BHC|5002|251^SPO2^BHC||||||F
OBX||CE|7^^BHC|5002|259^PR^BHC||||||F
OBX||CE|7^^BHC|5002|252^PI^BHC||||||F
OBX||CE|5^^BHC||5004^NIBP^BHC||||||F
OBX||CE|7^^BHC|5004|351^NIBP S^BHC||||||F
OBX||CE|7^^BHC|5004|352^NIBP D^BHC||||||F
OBX||CE|7^^BHC|5004|353^NIBP M^BHC||||||F
OBX||CE|7^^BHC|5004|354^NIBP SDP^BHC||||||F
OBX||CE|7^^BHC|5004|355^NIBP PR^BHC||||||F
OBX||CE|5^^BHC||5005^RESP^BHC||||||F
OBX||CE|7^^BHC|5005|401^RR^BHC||||||F
OBX||CE|5^^BHC||5012^CO^BHC||||||F
OBX||CE|7^^BHC|5012|751^C.O.^BHC||||||F
OBX||CE|7^^BHC|5012|752^C.I.^BHC||||||F
OBX||CE|7^^BHC|5012|753^TI^BHC||||||F
OBX||CE|7^^BHC|5012|754^TB^BHC||||||F
OBX||CE|5^^BHC||5031^IBP P1^BHC||||||F
OBX||CE|7^^BHC|5031|1501^Sys^BHC||||||F
OBX||CE|7^^BHC|5031|1502^Dia^BHC||||||F
OBX||CE|7^^BHC|5031|1503^Mean^BHC||||||F
OBX||CE|7^^BHC|5031|1504^PR^BHC||||||F
OBX||CE|7^^BHC|5031|1505^PPV^BHC||||||F
OBX||CE|5^^BHC||5039^ART^BHC||||||F
OBX||CE|7^^BHC|5039|1661^Sys^BHC||||||F
OBX||CE|7^^BHC|5039|1662^Dia^BHC||||||F
OBX||CE|7^^BHC|5039|1663^Mean^BHC||||||F
OBX||CE|7^^BHC|5039|1664^PR^BHC||||||F
OBX||CE|7^^BHC|5039|1665^PPV^BHC||||||F
OBX||CE|5^^BHC||5021^TEMPA^BHC||||||F
OBX||CE|7^^BHC|5021|1051^T01^BHC||||||F
OBX||CE|7^^BHC|5021|1052^T02^BHC||||||F

2. Send the supported parameter and parameter value data within the set time interval.

MSH|^~\&|||||20230713104819||ORU^R01^ORU_R01|1004|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F

OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||ST|4203^Blood^BHC||O||||||F
OBX||ST|4204^Pace^BHC||U||||||F
PV1||I|CCU^^&36&0&0||||||||||||||A
OBR||||CMS|||20230717170425
OBX||NM|201^HR^BHC|5001|60||||||F
OBX||NM|219^PVCs^BHC|5001|0||||||F
OBX||NM|220^QT^BHC|5001|353||||||F
OBX||NM|202^QTC^BHC|5001|353||||||F
OBX||NM|203^DELTAQTC^BHC|5001|0||||||F
OBX||NM|204ST-I^BHC|5001|-0.03||||||F
OBX||NM|205ST-II^BHC|5001|-0.02||||||F
OBX||NM|206ST-III^BHC|5001|0.03||||||F
OBX||NM|207ST-aVR^BHC|5001|0.02||||||F
OBX||NM|208ST-aVL^BHC|5001|-0.02||||||F
OBX||NM|209ST-aVF^BHC|5001|-0.01||||||F
OBX||NM|217ST-Va^BHC|5001|0.02||||||F
OBX||NM|218ST-Vb^BHC|5001|-0.03||||||F
OBX||NM|251^SPO2^BHC|5002|99||||||F
OBX||NM|252^PI^BHC|5002|7.46||||||F
OBX||NM|259^PR^BHC|5002|60||||||F
OBX||NM|351^NIBP S^BHC|5004|125|11^mmHg^BHC|||||F||APERIODIC|20230717165605
OBX||NM|352^NIBP D^BHC|5004|84|11^mmHg^BHC|||||F||APERIODIC|20230717165605
OBX||NM|353^NIBP M^BHC|5004|96|11^mmHg^BHC|||||F||APERIODIC|20230717165605
OBX||NM|355^NIBP PR^BHC|5004|60||||||F||APERIODIC|20230717165605
OBX||NM|401^RR^BHC|5005|20||||||F
OBX||NM|751^C.O.^BHC|5012|5.2||||||F||APERIODIC|20230717162612
OBX||NM|752^C.I.^BHC|5012|2.9||||||F||APERIODIC|20230717162612
OBX||NM|754^TB^BHC|5012|36.89|21^C^BHC|||||F||APERIODIC|20230717162612
OBX||NM|1501^Sys^BHC|5031|122|11^mmHg^BHC|||||F
OBX||NM|1502^Dia^BHC|5031|81|11^mmHg^BHC|||||F
OBX||NM|1503^Mean^BHC|5031|99|11^mmHg^BHC|||||F
OBX||NM|1504^PR^BHC|5031|60||||||F
OBX||NM|1505^PPV^BHC|5031|2||||||F
OBX||NM|1521^Sys^BHC|5032|122|11^mmHg^BHC|||||F
OBX||NM|1522^Dia^BHC|5032|81|11^mmHg^BHC|||||F
OBX||NM|1523^Mean^BHC|5032|99|11^mmHg^BHC|||||F
OBX||NM|1524^PR^BHC|5032|60||||||F
OBX||NM|1525^PPV^BHC|5032|2||||||F
OBX||NM|2151^Drop Speed^BHC|5060|50||||||F
OBX||NM|2152^Flowing^BHC|5060|150||||||F
OBX||NM|1051^T01^BHC|5021|36.6|21^C^BHC|||||F
OBX||NM|1052^T02^BHC|5021|36.5|21^C^BHC|||||F

OBX||NM|2052^Score Type^BHC|5058|0|||||F
OBX||ST|4002^Monitor Name^BHC||36_|||||F
OBX||CE|4021^Tec Highest Level^BHC||0^Unknown^BHC|||||F
OBX||CE|4022^Phy Highest Level^BHC||3^Middle Level^BHC|||||F
OBX||CE|4023^Alarm Setting^BHC||0^Normal^BHC|||||F
OBX||CE|4024^ECG Lead Type^BHC||3^6 Lead^BHC|||||F
OBX||CE|4025^PR_Source^BHC||1^SpO2^BHC|||||F
OBX||CE|4026^RR_Source^BHC||1^ECG^BHC|||||F

3．Data completed by NIBP aperiodic measurement

MSH|^~\&||||20230518101324||ORU^R01^ORU_R01|1005|P|2.6|||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F
OBX||NM|4203^Blood^BHC||3|||||F
OBX||NM|4204^Pace^BHC||2|||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518101324
OBX||NM|351^NIBP S^BHC|5004|125|11^mmHg^BHC||||F||APERIODIC|20230518101323
OBX||NM|352^NIBP D^BHC|5004|84|11^mmHg^BHC||||F||APERIODIC|20230518101323
OBX||NM|353^NIBP M^BHC|5004|96|11^mmHg^BHC||||F||APERIODIC|20230518101323
OBX||NM|354^NIBP SDP^BHC|5004|41|11^mmHg^BHC||||F||APERIODIC|20230518101323
OBX||NM|355^NIBP PR^BHC|5004|60|||||F||APERIODIC|20230518101323
OBX||ST|4002^Monitor Name^BHC||Monitor|||||F
OBX||CE|4021^Tec Highest Level^BHC||1^Tips^BHC|||||F
OBX||CE|4022^Phy Highest Level^BHC||5^None^BHC|||||F
OBX||CE|4023^Alarm Setting^BHC||1^Pause^^BHC|||||F

4．C.O. aperiodic parameter measurement completed data

MSH|^~\&||||20230518101323||ORU^R01^ORU_R01|1006|P|2.6|||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F
OBX||NM|4203^Blood^BHC||3|||||F
OBX||NM|4204^Pace^BHC||2|||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518101323
OBX||NM|751^C.O.^BHC|5012|5.2|||||F||APERIODIC|20230518101323
OBX||NM|752^C.I.^BHC|5012|2.9|||||F||APERIODIC|20230518101323
OBX||NM|753^TI^BHC|5012|-1|||||F||APERIODIC|20230518101323
OBX||NM|754^TB^BHC|5012|36.87|21^C^BHC||||F||APERIODIC|20230518101323
OBX||ST|4002^Monitor Name^BHC||Monitor|||||F
OBX||CE|4021^Tec Highest Level^BHC||1^Tips^BHC|||||F

OBX||CE|4022^Phy Highest Level^BHC||5^None^BHC||||||F
OBX||CE|4023^Alarm Setting^BHC||1^Pause^BHC||||||F
OBX||CE|4024^ECG Lead Type^BHC||1^3 Lead^BHC||||||F
OBX||CE|4025^PR_Source^BHC||1^SpO2^BHC||||||F
OBX||CE|4026^RR_Source^BHC||1^ECG^BHC||||||F

5. NMT aperiodic parameter measurement completed data

MSH|^~\&||||20230518115149||ORU^R01^ORU_R01|1025|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518115149
OBX||ST|2201^Stimulate Current^BHC|5057|25^25^25^25||||||F
||APERIODIC|20230518115149
OBX||ST|2202^Measure Current^BHC|5057|24.80^24.81^24.80^24.79||||||F
||APERIODIC|20230518115149
OBX||ST|2203^Bargraph Strength^BHC|5057|70^10^10^0||||||F
||APERIODIC|20230518115149
OBX||NM|2204^Measure Result^BHC|5057|3||||||F||APERIODIC|20230518115149
OBX||NM|2205^Result Type^BHC|5057|0||||||F||APERIODIC|20230518115149
OBX||NM|2206^Stimulate Cycle^BHC|5057|-1||||||F||APERIODIC|20230518115149
OBX||NM|2207^Stimulate Mode^BHC|5057|3||||||F||APERIODIC|20230518115149
OBX||NM|2208^Auto Mode^BHC|5057|-1||||||F||APERIODIC|20230518115149
OBX||NM|2209^Bargraph Number^BHC|5057|4||||||F||APERIODIC|20230518115149
OBX||ST|4002^Monitor Name^BHC||Monitor||||||F
OBX||CE|4021^Tec Highest Level^BHC||1^Tips^BHC||||||F
OBX||CE|4022^Phy Highest Level^BHC||3^Middle Level^BHC||||||F
OBX||CE|4023^Alarm Setting^BHC||0^Normal^BHC||||||F
OBX||CE|4024^ECG Lead Type^BHC||1^3 Lead^BHC||||||F
OBX||CE|4025^PR_Source^BHC||1^SpO2^BHC||||||F
OBX||CE|4026^RR_Source^BHC||1^ECG^BHC||||||F

6. PAWP aperiodic parameter measurement completed data

MSH|^~\&||||20230518114729||ORU^R01^ORU_R01|1026|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F

PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518114729
OBX||NM|1706^Pawp^BHC|5041|81||||||F||APERIODIC|20230518114729
OBX||ST|4002^Monitor Name^BHC||Monitor||||||F
OBX||CE|4021^Tec Highest Level^BHC||1^Tips^BHC||||||F
OBX||CE|4022^Phy Highest Level^BHC||5^None^BHC||||||F
OBX||CE|4023^Alarm Setting^BHC||0^Normal^BHC||||||F
OBX||CE|4024^ECG Lead Type^BHC||1^3 Lead^BHC||||||F
OBX||CE|4025^PR_Source^BHC||1^SpO2^BHC||||||F
OBX||CE|4026^RR_Source^BHC||1^ECG^BHC||||||F

7. EWS aperiodic parameter measurement completed data

MSH|^~\&||||20230518135302||ORU^R01^ORU_R01|1027|P|2.6|||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518135302
OBX||NM|2051^Total Score^BHC|5058|5||||||F||APERIODIC|20230518135302
OBX||NM|2052^Score Type^BHC|5058|0||||||F||APERIODIC|20230518135302
OBX||ST|4002^Monitor Name^BHC||Monitor||||||F
OBX||CE|4021^Tec Highest Level^BHC||1^Tips^BHC||||||F
OBX||CE|4022^Phy Highest Level^BHC||1^Tips^BHC||||||F
OBX||CE|4023^Alarm Setting^BHC||0^Normal^BHC||||||F
OBX||CE|4024^ECG Lead Type^BHC||1^3 Lead^BHC||||||F
OBX||CE|4025^PR_Source^BHC||1^SpO2^BHC||||||F
OBX||CE|4026^RR_Source^BHC||1^ECG^BHC||||||F

8. GCS aperiodic parameter measurement completed data

MSH|^~\&||||20230518135251||ORU^R01^ORU_R01|1028|P|2.6|||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518135251
OBX||NM|2101^Total Score^BHC|5059|9||||||F||APERIODIC|20230518135251
OBX||NM|2102^Eye Opening^BHC|5059|2||||||F||APERIODIC|20230518135251
OBX||NM|2103^Verbal Response^BHC|5059|3||||||F||APERIODIC|20230518135251

OBX||NM|2104^Motor Response^BHC|5059|4||||||F||APERIODIC|20230518135251
OBX||ST|4002^Monitor Name^BHC||Monitor||||||F
OBX||CE|4021^Tec Highest Level^BHC||1^Tips^BHC||||||F
OBX||CE|4022^Phy Highest Level^BHC||5^None^BHC||||||F
OBX||CE|4023^Alarm Setting^BHC||0^Normal^BHC||||||F
OBX||CE|4024^ECG Lead Type^BHC||1^3 Lead^BHC||||||F
OBX||CE|4025^PR_Source^BHC||1^SpO2^BHC||||||F
OBX||CE|4026^RR_Source^BHC||1^ECG^BHC||||||F

9. Parameter alarm upper and lower limits

MSH|^~\&|||||20230713104818||ORU^R01^ORU_R01|1009|P|2.6||||||UTF-8
PID|||||||M
OBX||NM|4201^Height^BHC||66.9|32^inch^BHC|||||F
OBX||NM|4202^Weight^BHC||143.3|42^lb^BHC|||||F
OBX||NM|4203^Blood^BHC||4||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230713104818
OBX||NM|1^^BHC|201|348||||||F
OBX||NM|2^^BHC|201|347||||||F
OBX||NM|1^^BHC|251|100||||||F
OBX||NM|2^^BHC|251|90||||||F
OBX||NM|1^^BHC|351|120|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|351|70|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|352|70|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|352|40|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|353|90|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|353|50|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|354|90|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|354|50|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|801|49|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|801|20|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|802|3|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|803|30||||||F
OBX||NM|2^^BHC|803|8||||||F
OBX||NM|1^^BHC|881|3.0||||||F
OBX||NM|2^^BHC|881|0.5||||||F
OBX||NM|1^^BHC|882|2.0||||||F
OBX||NM|2^^BHC|882|0.5||||||F
OBX||NM|1^^BHC|841|100|14^%^BHC|||||F
OBX||NM|2^^BHC|841|18|14^%^BHC|||||F
OBX||NM|1^^BHC|842|100|14^%^BHC|||||F
OBX||NM|2^^BHC|842|18|14^%^BHC|||||F

OBX||NM|1^^BHC|821|82|||||F
OBX||NM|2^^BHC|821|0|||||F
OBX||NM|1^^BHC|822|77|||||F
OBX||NM|2^^BHC|822|0|||||F
OBX||NM|1^^BHC|401|30|||||F
OBX||NM|2^^BHC|401|8|||||F
OBX||NM|1^^BHC|1501|120|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|1501|70|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|1502|70|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|1502|40|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|1503|90|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|1503|50|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|1521|120|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|1521|70|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|1522|70|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|1522|40|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|1523|90|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|1523|50|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|1051|39.0|21^C^BHC|||||F
OBX||NM|2^^BHC|1051|36.0|21^C^BHC|||||F
OBX||NM|1^^BHC|1102|39.0|21^C^BHC|||||F
OBX||NM|2^^BHC|1102|36.0|21^C^BHC|||||F

10. Parameter alarm switch
MSH|^~\&|||||20230518101318||ORU^R01^ORU_R01|1011|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518101318
OBX||NM|4^^BHC|201|1||||||F
OBX||NM|4^^BHC|251|1||||||F
OBX||NM|4^^BHC|351|1||||||F
OBX||NM|4^^BHC|352|1||||||F
OBX||NM|4^^BHC|353|1||||||F
OBX||NM|4^^BHC|354|1||||||F
OBX||NM|4^^BHC|401|1||||||F
OBX||NM|4^^BHC|754|1||||||F
OBX||NM|4^^BHC|1501|1||||||F
OBX||NM|4^^BHC|1502|1||||||F
OBX||NM|4^^BHC|1503|1||||||F
OBX||NM|4^^BHC|1521|1||||||F

OBX||NM|4^^BHC|1522|1||||||F
OBX||NM|4^^BHC|1523|1||||||F
OBX||NM|4^^BHC|1051|1||||||F
OBX||NM|4^^BHC|1052|1||||||F

11. Parameter alarm level
MSH|^~\&|||||20230518101318||ORU^R01^ORU_R01|1010|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230518101318
OBX||NM|3^^BHC|201|2||||||F
OBX||NM|3^^BHC|251|2||||||F
OBX||NM|3^^BHC|351|2||||||F
OBX||NM|3^^BHC|352|2||||||F
OBX||NM|3^^BHC|353|2||||||F
OBX||NM|3^^BHC|354|2||||||F
OBX||NM|3^^BHC|401|2||||||F
OBX||NM|3^^BHC|754|2||||||F
OBX||NM|3^^BHC|1501|2||||||F
OBX||NM|3^^BHC|1502|2||||||F
OBX||NM|3^^BHC|1503|2||||||F
OBX||NM|3^^BHC|1521|2||||||F
OBX||NM|3^^BHC|1522|2||||||F
OBX||NM|3^^BHC|1523|2||||||F
OBX||NM|3^^BHC|1051|2||||||F
OBX||NM|3^^BHC|1052|2||||||F

4.3 Waveform Data Sending

4.3.1 Message content

The waveform message consists of four parts: waveform channel definition data (CHN), timing information (time and date observed), actual waveform data (WAV) and waveform annotation (ANO). The waveform data of each module can be either single channel or multi-channel, and the data of each channel can use separate timing information, channel definition data, digital waveform data and waveform annotation. In addition, depending on the situation, the waveform data channels of multiple sensors can share timing information, channel definition data, and waveform annotations.

4.3.2 Waveform sending form

The waveform data is sent by the ORU_R01 message, which contains waveform specific data and channel attributes (sensitivity, sample rate, etc.). Both single and multi-channel waveforms can be sent.

In addition, the sampling rate sent by the ECG waveform can be selected, which is set in the HL7 interface.

4.3.3 Format

```
MSH   Message Header
PID   Patient ID Segment
OBX   Patient Height
OBX   Patient Weight
OBX   Patient Blood
OBX   Patient Paced
PV1   Patient location information segment
OBX   Channel specification data (channel identifier,
      channel sensitivity, sample rate, Max/min)
OBX   Timing information (time and date of observation)
OBX   Waveform data (actual waveform data)
OBX   Waveform annotation
OBX   Channel specification data (channel identifier,
      channel sensitivity, sample rate, Max/min)
OBX   Timing information (time and date of observation)
OBX   Waveform data (actual waveform data)
OBX   Waveform annotation
......
```

4.3.4 Definition of message segment

(1) MSH Segment

In this message, the message type is: ORU_R01; Message control ID: 1015, the control ID is defined in Appendix *A.5 Controls and Settings*.

(2) PID segment, patient observation OBX segment, PV1 segment

Same as above.

(3) OBX Segment

◆ OBX channel description data

SEQ	LEN	DT	OPT	Field Name	Description
2	3	ID	O	Value Type	CD
3	705	CWE	R	Observation Identifier	Waveform identifier， “30001ST-I^BHC” indicates the channel definition for sending

					the waveform of ST_I
4	20	ST	O	Observation Sub-ID	Waveform channel ID, for example, 1 indicates channel 1
5	99999	CD	O	Observation Results	Channel definition
11	1	ID	R	Observation Results Status	F

- ✧ Channel definition: < Channel Identifier (WVI) >^< Waveform Source (WVS) >^< Channel sensitivity/unit (CSU) >^< Channel annotation Parameter (CCP) >^< Channel sampling frequency (NM) > ^< Minimum/Maximum (NR) >;
- ✧ Channel identifier sub-component: < Channel Number (NM)> &< Channel name (ST)>
- ✧ Waveform source sub-components: <Source Name 1(ST)> &< Source Name 2 (ST) >
- ✧ Channel sensitivity/Unit subcomponent: <Channel sensitivity (NM)> &<Unit of Measurement Identifier (ST)>&< Description of Unit of measurement (ST)>&<Unit of Measurement Code System (ID)>&<Alternate Unit of Measurement Identifier (ST)>&<Description of Alternate Units of Measurement (ST)>&<Alternate Unit of Measurement Code System (ID)>
- ✧ Sub-components of channel labeling parameters: <Channel scale sensitivity correction factor (NM)>&<Channel scale baseline (NM)> &< Channel scale time skewness (NM)>
- ✧ Minimum/Maximum subcomponent: <Minimum value (NM)>&<Maximum value (NM)>

◆ OBX Timing information

SEQ	LEN	DT	OPT	Field Name	Description
2	3	ID	O	Value Type	TS
3	705	CWE	R	Observation Identifier	Waveform identifier, “30001ST-I^BHC” Indicates that the message sent is the waveform observation date and time of ST_I
4	20	ST	O	Observation Sub-ID	Waveform channel ID, for example, 1 indicates channel 1
5	65536	CD	O	Observation Results	Observe date and time YYYYMMddHHmmss
11	1	ID	R	Observation Results Status	F

- ✧ The timing information section determines the date and time of the first data point of the child ID observed for a particular waveform channel group

◆ Waveform data

SEQ	LEN	DT	OPT	Field Name	Description

2	3	ID	O	Value Type	NM or NA
3	705	CWE	R	Observation Identifier	Waveform parameter name identifier, “30001ST-I^BHC”Represents the waveform data of ST_I
4	20	ST	O	Observation Sub-ID	Waveform channel ID, for example, 1 indicates channel 1
5	65536	CD	O	Observation Results	Waveform data, single channel or multi-channel waveform data, the length is not greater than 65536
11	1	ID	R	Observation Results Status	F

Waveform data:

- ✧ One-dimensional array: <Waveform value 1>^<Waveform value 2>^<Waveform value 3>... (NA)
- ✧ Two-dimensional array: <Waveform value 11>^<Waveform value 12>^<Waveform value 13>... ~< Waveform value 21>^<Waveform value 22>^<Waveform value 23>... (NA)
- ✧ Multi-channel: <Channel 1 data 1> ^<Channel 2 data 1>^<Channel 3 data 1>... ~ <Channel 1 data 2> ^<Channel 2 data 2>^<Channel 3 data 2>... ~ (NM)

◆ Waveform Notes

SEQ	LEN	DT	OPT	Field Name	Description
2	3	ID	O	Value Type	CE
3	705	CWE	R	Observation Identifier	Waveform parameter name identifier，“30001ST-I^BHC”Waveform annotation representing ST_I
4	20	ST	O	Observation Sub-ID	Waveform channel ID, for example, 1 indicates channel 1
5	65536	CD	O	Observation Results	Waveform annotation content
11	1	ID	R	Observation Results Status	F

4.3.5 Example

```
MSH|^~\&|||||20230518135303||ORU^R01^ORU_R01|1015|P|2.6|||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
```

OBR||||Monitor|||20230518135303
OBX||CD|30002^ECG_II^BHC|1|1^^0.001&mV^^256^-300000&300000||||||F
OBX||TS|30002^ECG_II^BHC|1|20230614102744||||||F
OBX||NA|30002^ECG_II^BHC|1|-187^-187^-193^-194^-187^-184^-190^-196^-168^-72^45^247
^456^625^740^828^895^939^958^951^921^887^814^708^590^454^278^66^-141^-287^-344^-3
36^-310^-280^-267^-281^-310^-328^-321^-296^-281^-285^-288^-277^-264^-263^-271^-273^-2
60^-253^-315^-602^-1177^-1682^-1949^1257^359^2434^4536^6444^7986^8625^
7919^6124^4423^2039^-223^-2071^-2998^-2814^--1986^-1138^-583^-359^-369^-548^-756^-76
9^-586^-424^-458^-616^-668^-557^-435^-426^-522^-587^-518^-394^-350^-405^-430^-360^-24
8^-192^-212^-246^-203^-104^-35^-28^-29^25^119^191^211^212^249^331^418^467^495^
548^632^713^790^863^1003^1175^1328^1437^1517^1598^1658^1648^1552^1418^1332^1260^
1222^1179^1068^845^530^181^-131^-367^-494^-619^-706^-754^-770^-765^-747^-718^-709^-7
21^-732^-724^-699^-677^-676^-682^-671^-648^-626^-622^-625^-625^-621^-610^-601^-597^-5
86^-566^-549^-540^-540^-543^-545^-545^-536^-523^-522^-526^-523^-517^-515^-513^-510^-5
05^-490^-480^-466^-455^-447^-446^-445^-442^-435^-422^-412^-410^-419^-423^-418^-410^-4
03^-395^-386^-381^-383^-393^-393^-376^-357^-347^-345^-343^-343^-351^-348^-338^-329^-3
30^-333^-334^-331^-326^-323^-313^-302^-296^-291^-286^-284^-286^-287^-288^-294^-288^-2
71^-256^-252^-265^-277^-280^-274^-260^-242^-235^-238^-236^-228^-218^-213^-222^-230^-2
34^-229^-215^-205^-196^-196^-201^-203^-197^-191^-193^-204^-210^-205^-197^-164^-66^53^
256^465^630^746^831^898^943^961^958^926^875^780^679^578^446^264^59^-130^-265^-326
^-331^-316^-297^-288^-286^-285^-281||||||F
OBX||CE|30002^ECG_II^BHC|1|||||||F
OBX||CD|30082^IBP_P2^BHC|1|1^^0.5&mmHg^^100^-5000&36000||||||F
OBX||TS|30082^IBP_P2^BHC|1|20230518135303||||||F
OBX||NA|30082^IBP_P2^BHC|1|9099^9429^9746^10066^10391^10709^11010^11273^11477^1
1633^11768^11889^11991^12076^12149^12200^12222^12222^12205^12167^12114^12050^11
969^11875^11769^11646^11510^11373^11236^11088^10918^10728^10537^10374^10227^100
71^9915^9799^9750^9751^9759^9756^9758^9773^9784^9781^9767^9742^9708^9664^9618^95
84^9562^9536^9502^9468^9437^9406^9377^9347^9310^9275^9248^9219^9182^9146^9117^90
89^9055^9021^8992^8962^8922^8882^8849^8818^8789^8767^8737^8688^8642^8618^8601^85
69^8530^8498^8473^8442^8402^8364^8330^8291^8238^8181^8146^8154^8215^8355^8596^89
05^9230^9549^9871^10200^10529||||||F
OBX||CE|30082^IBP_P2^BHC|1|||||||F

5 Query Request Interface

This section introduces the relevant content of the query interface.

5.1 Communication process

5.1.1 Network Connection

The query interface is only used as the server to wait for connection requests from the client, and the process of establishing a connection is the same as actively sending the interface as the server. Therefore, please refer to section *4.1.1*.

5.1.2 Network Connection Settings

Query interface as server: The client needs to know the IP and port number of the server in order to connect. The IP address of the server can be viewed by clicking on the [Main menu] button of the monitor and clicking on the [System Info]. The port number is fixed to 6202.

5.1.3 Data Interaction

The query interface adopts a communication mode of request and response, which means that the server will not send any data to the client until it receives the query message (QRY_R02) sent by the client.

The client can query the data of one, multiple, or all monitors at once, and can specify the type of data to be queried. As the query interface of the server, after receiving the query message, it will first return an ACK message as a response, and finally return a message containing the query result (ORF_R04). After receiving the result returned by the interface, the client will send an ACK or NCK as a response, but this response will be ignored. If the server does not specify the data type to query, the interface will not return any messages.

If the client sends a query message followed by a new query message, the interface will first respond to the previous query message, and then send an ACK message warning the client that the query message interval is too short. Therefore, when sending query messages, it is important for the client to keep track of the time interval and wait at least 10 seconds before sending new query messages.

5.1.4 Send Content

The following data is sent after the client sends the correct query message.

1. Module parameter values.
2. Physiological alarm data.

3. Technical alarm data.

4. Module configuration and device status include: device standby state, physiological and technical highest alarm level, device alarm state, ECG lead type, PR source, RR source.

5. Module parameter alarm configuration data, including alarm switch, alarm level, and alarm upper and lower limit values.

5.2 QRY_R02 Message

QRY_ The QRY in R02 represents the query observation results, and R02 represents the event. The query message type sent by the client is QRY_ R02 generally forms a query and response mode with ORF.

QRY_ R02 is divided into three parts, namely Message Header (MSH), Query Definition (QRD), and Query Filter (QRF).

5.2.1 Format

MSH	Message header
QRD	Query Definition
QRF	Query filtering

5.2.2 Example

MSH|^~\&|||||20220424095857||ORY^R02^ORY_R02|1029|P|2.6||||||UTF-8
QRD|20220424095857|R|I|2|||||RES
QRF|MON||||0&0^1^0^0

5.2.3 MSH Segment

The MSH segment is the first segment of an HL7 message, which defines the message type.

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	1	ST	R	No	Field separator	|
2	4	ST	R	No	Code Characters	^~\&
9	15	MSG	R	No	Date/time of the message	ORY^R02^ORY_R02
7	24	DTM	R	No	Message Type	yyyyMMddHHmmss
10	199	ST	R	No	Message Control ID	Contains a number or other identifier that uniquely identifies the message, such as MSH | ^~&| |

						| | 20220424095857 | | ORY ^ R02 ^ORY_R02| 1029 | P| 2.6|||||UTF-8 1029 represents the control ID of the real-time query message, indicating that this HL7 message is a real-time query message. Please refer to *Appendix A.5 controls and settings*
11	3	PT	R	No	Process ID	P
12	60	VID	R	No	Version ID	2.6

5.2.4 QRD Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	24	DTM	R	No	Query Data/Time	Format:yyyyMMddHHmmss
2	1	ID	R	No	Query	Must always be R
3	1	ID	R	No	Query Priority	Must always be I
4	10	ST	R	No	Query ID	Cannot be empty and must be less than 16 characters
9	250	CWE	R	Yes	Filter themes	Must be a result between query and response RES

Example:QRD|20220424095857|R|I|2|||||RES

Message processing rules:

- The first field must be in the correct time format, otherwise the interface will return an error message.
- The second field must be R, otherwise the interface will return an error message.
- The third field must be I, otherwise the interface will return an error message.
- The fourth field cannot be empty, otherwise the interface will return an error message, and the interface cannot determine the uniqueness of the ID

5.2.5 QRF Segment

QRF defines the detailed content to be queried, and there is only one query condition (QRF) at a time.

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	20	ST	R	Yes	Filter the location of	In the monitoring data of queries and responses, it must always be a

					the theme	MON
5	60	ST	O	Yes	Other QRY topic filtering	Contains locally defined filtering criteria for use between two systems, as detailed in *5.2.5.1*

Example: QRF | MON | | | | 0&0 ^ 1 ^ 0 ^ 0, where 1 in the segment indicates that the query data type is physiological parameter data.

Message processing rules:

- The first field must be MON, otherwise the interface will return an error message
- The fifth field cannot be empty, otherwise the interface will return an error message

5.2.5.1 Field 5 of QRF

The format for querying and filtering topics is as follows:

<IP>&<IPSeq>^<SendType>^ 0 ^ 0

Please refer to the table below for the description of all components in this field

Composition	Data type	Numerical range	describe
<IP>	integer	0-4294967295	Because CMS and PDS gateways are connected to multiple bedside monitors simultaneously, IP addresses are used to identify unique bedside monitors or telemetry monitors. 0 represents requesting data from the monitor and does not require an IP address
<IpSeq>	integer	0-255	Subtract 1 from the serial number of the telemetry transmitter, which is used when connecting to the central monitoring system. 0 represents requesting data from the monitor and does not require an IP address. Note: This is not a port number
<SendType>	integer	1-63	Requested data type When data is represented in binary Physiological parameter data: The 0th value is 1; Physiological alarm data: The first value is 1; Technical alarm data: The second value is 1; Alarm setting data: The third digit value is 1; Other configuration data: The fourth digit value is 1; *Waveform data: The 5th digit value is 1;* All data: digits 0-5 are 1

5.3 ACK Message

5.3.1 Format

ACK message is the error message returned by the interface upon receiving an incorrect query message from the client. The client can correct its own query message based on this error message to obtain the desired result message.

5.3.2 Example

MSH|^~\\&|||||20230522165958||ACK|1018|P|2.6||||||UTF-8
MSA|AR|1029|The request is too fast.

5.3.3 MSH Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	1	ST	R	No	Field separator	|
2	4	ST	R	No	Code Characters	^~\&
9	15	MSG	R	No	Date/time of the message	ACK
7	24	DTM	R	No	Message Type	yyyyMMddHHmmss
10	199	ST	R	No	Message Control ID	1018：Control ID of ACK confirmation message
11	3	PT	R	No	Process ID	P
12	60	VID	R	No	Version ID	2.6

MSH|^~\\&|||||20230522165958||ACK|1018|P|2.6||||||UTF-8
The message type field for this segment is: ACK

5.3.4 MSA Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	2	ID	R	No	Confirmation Code	AA: Initial mode: Application acceptance; AE: initial mode: application Software bug; AR: Initial mode: Application rejection;

						CA: Application confirmation in enhanced mode, indicating acceptance; CE: Acceptance confirmation in enhanced mode, indicating an error; CR: Acceptance confirmation in enhanced mode, indicating rejection.
2	199	ST	R	No	Message Control ID	Consistent with the control ID of the query message
3	80	ST	O	No	text message	Describe the reason for the error or rejection, such as: The request is too fast.

MSA|AR|1029|The request is too fast.

5.3.5 ERR Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	493	ELD	O	Yes	Error codes and locations	Error codes and locations
2	18	ERL	O	Yes	error location	
3	705	CWE	R	No	HL7 error code	0：Received message
4	2	ID	R	No	Severity	W： Warning I： Information E： Error F： Fatal Error
5	705	CWE	O	No	Application Software bug Code	1：Not connected 2：Unauthorized
6	80	ST	O	Yes	Application Software bug parameter	<IP>,<IPSeq> For details, refer to section 5.2.5.1

5.4 ORF_R04 Message

QRY_ The QRY in R04 represents the query observation results, while R04 represents the event. The query result message type sent by the server is QRY_ R04.

5.4.1 Format

Patient_1	MSH Segment	MSA Segment	QRD Segment	QRF Segment	PID Segment	OBR Segment	OBX Segment	...

Patient_2	MSH Segment	MSA Segment	QRD Segment	QRF Segment	PID Segment	OBR Segment	OBX Segment	...

.....

Patient_n	MSH Segment	MSA Segment	QRD Segment	QRF Segment	PID Segment	OBR Segment	OBX Segment	...

5.4.2 Example

```
MSH|^~\\&||||20230522165958||ORF^R04^ORF_R04|1030|P|2.6|||||UTF-8
MSA|AA|1029
QRD|20230522170331|R|I|2|||||RES
QRF|MON||||0&0^1^0^0
PID|||2||Mis^Lucy^Bob||19951108|F
OBR||||Monitor|||20230522165958
OBX||NM|201^HR^BHC|5001|-1||||||F
OBX||NM|219^PVCs^BHC|5001|-1||||||F
OBX||NM|251^SPO2^BHC|5002|99||||||F
OBX||NM|259^PR^BHC|5002|60||||||F
OBX||NM|252^PI^BHC|5002|7.45||||||F
OBX||NM|258^RR^BHC|5002|20||||||F
OBX||NM|351^NIBP S^BHC|5004|-1||||||F
OBX||NM|352^NIBP D^BHC|5004|-1||||||F
OBX||NM|353^NIBP M^BHC|5004|-1||||||F
OBX||NM|354^NIBP SDP^BHC|5004|-1||||||F
OBX||NM|355^NIBP PR^BHC|5004|-1||||||F
OBX||NM|401^RR^BHC|5005|20||||||F
OBX||NM|751^C.O.^BHC|5012|-1||||||F
OBX||NM|752^C.I.^BHC|5012|-1||||||F
OBX||NM|753^TI^BHC|5012|-1||||||F
OBX||NM|754^TB^BHC|5012|36.98||||||F
OBX||NM|1501^Sys^BHC|5031|122||||||F
OBX||NM|1502^Dia^BHC|5031|81||||||F
OBX||NM|1503^Mean^BHC|5031|99||||||F
OBX||NM|1504^PR^BHC|5031|60||||||F
OBX||NM|1505^PPV^BHC|5031|-1||||||F
```

OBX||NM|1521^Sys^BHC|5032|122||||||F
OBX||NM|1522^Dia^BHC|5032|81||||||F
OBX||NM|1523^Mean^BHC|5032|99||||||F
OBX||NM|1524^PR^BHC|5032|60||||||F
OBX||NM|1525^PPV^BHC|5032|-1||||||F
OBX||NM|1051^T01^BHC|5021|36.6||||||F
OBX||NM|1052^T02^BHC|5021|36.5||||||F
OBX||NM|2151^Drop Speed^BHC|5060|49||||||F
OBX||NM|2152^Flowing^BHC|5060|147||||||F
OBX||NM|2051^Total Score^BHC|5058|-1||||||F
OBX||NM|2052^Score Type^BHC|5058|0||||||F
OBX||NM|2101^Total Score^BHC|5059|-1||||||F
OBX||NM|2102^Eye Opening^BHC|5059|-1||||||F
OBX||NM|2103^Verbal Response^BHC|5059|-1||||||F
OBX||NM|2104^Motor Response^BHC|5059|-1||||||F

5.4.3 MSH Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	1	ST	R	No	Field separator	|
2	4	ST	R	No	Code Characters	^~\&
9	15	MSG	R	No	Date/time of the message	ORF^R04^ORF_R04
7	24	DTM	R	No	Message Type	yyyyMMddHHmmss
10	199	ST	R	No	Message Control ID	Contains a number or other identifier that uniquely identifies the message, For example: MSH | ^~ &| | | 20230522165958 | | ORF ^ R04 ^ ORF_ R04 | 1030 | P | 2.6|||||UTF-8 1030 represents the control ID of the query result message type, indicating that this HL7 message is the returned query result message. Please refer to *Appendix A.5 Controls and settings*
11	3	PT	R	No	Process ID	P
12	60	VID	R	No	Version ID	2.6

5.4.4 MSA Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	2	ID	R	No	Confirmation Code	AA: Initial mode: Application acceptance; AE: initial mode: application Software bug; AR: Initial mode: Application rejection; CA: Application confirmation in enhanced mode, indicating acceptance; CE: Acceptance confirmation in enhanced mode, indicating an error; CR: Acceptance confirmation in enhanced mode, indicating rejection.
2	199	ST	O	No	Message Control ID	1029 (consistent with the control ID of real-time query messages)
3	80	ST	O	No	text message	Describe the reason for the error or rejection, such as: The request is too fast.

5.4.5 QRD Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	24	DTM	R	No	Query date/time	Format：yyyyMMddHHmmss
2	1	ID	R	No	Query time codes	Must always beR
3	1	ID	R	Yes	Query Priority	Must always beI
4	10	ST	R	Yes	Query ID	Cannot be empty and must be less than 16 characters
9	250	CWE	R	No	Filter themes	Must be the result RES between query and response

5.4.6 QRF Segment

Field	Length	Data	Wash	Repeat	Field Name	Describe

		Type	ability	ability		
1	20	ST	R	Yes	Filter the location of the theme	In the monitoring data of queries and responses, it must always be a MON
5	60	ST	O	Yes	Other QRY topic filtering	For details, refer to section *5.2.5.1*

5.4.7 PID Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
3	250	CX	R	Yes	Patient identifier table	Patient's medical record number
5	250	XPN	R	Yes	Patient Name	First Name ^ Middle name ^ Last Name
7	24	DTM	O	No	Date/Time of Birth	YYYYMMDD
8	1	IS	O	No	Gender	M: Male F: Women U: Unknown

5.4.8 OBR Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
4	705	CWE	R	No	Universal Service Identifier	Monitor
7	24	DTM	O	No	Observation date/time	yyyyMMddHHmmss For example: OBR Monitor 20210914161435 Indicating observation time: September 14, 2021 16:14:35

5.4.9 OBX Segment

The definition and usage of OBX segment here are the same as section 4.2.7, please refer to section *4.2.7* for details.

6 Real time query interface

The difference between real-time interfaces and other interfaces is that only one patient's data is sent in a TCP connection. The patient's personal information does not appear in every HL7 message (that is, PID and PV1 segments do not appear in each HL7 message), and is only sent after the connection is established or changed. In addition, this interface also sets up UDP broadcast information to send machine system information and patient personal information.

6.1 Communication process

6.1.1 Network Connection

The method and process for establishing a connection between the client and server of the real-time query interface are the same as those for actively sending the interface, so we can refer to *4.1.1*.

6.1.2 Network Connection Settings

Real time query interface as the server: The client needs to know the IP and port number of the server in order to connect. The IP address of the server can be viewed by clicking on the [Main menu] button of the monitor and clicking on the [System Info]. The port number is fixed to 6210. In addition, the real-time query interface is also set with UDP broadcast information, which will send broadcast information to all hosts in the subnet where the monitor is located, with port number 6234.

6.1.3 Data Interaction

The real-time query interface adopts a communication mode that combines request and response with active sending. That is, after the client establishes a connection with the interface, the interface first sends online module information to the client, and then sends heartbeat packets every second. At the same time, the client needs to send a heartbeat packet to the interface. If the interface does not receive the heartbeat packet sent by the client within 10 seconds after establishing the connection, it will actively disconnect.

During the process of maintaining network connectivity, the client can send a message that needs to be queried (QRY_R02) and specify a response time interval. After receiving the correct query message, the interface will reply to the corresponding (ORU_R01) result message based on the time interval.

If the client sends a query and message followed by a new and different query message, the interface returns the result message of the new query message first, and then the previous result message. Each subsequent result message will also be sent in this way.

The real-time query interface also sets up UDP broadcast information, sending machine system information and patient personal information every second to inform other hosts on the network of its existence.

6.1.4 Send Content

6. Online broadcast information: including patient name, gender, doctor ID, date of birth, department, bed number, and monitor name;
7. TCP heartbeat packet information: sent once every 1 second;
8. Patient information: When the TCP connection is successfully established, the latest patient information will be sent to the client every second;
9. Response message: After receiving the query message, the interface will return the corresponding response message;
10. Physiological parameters: When the interface receives a query message as a physiological parameter, it sends physiological parameters according to the time interval;
11. Physiological alarm information: When the interface receives a query message as a physiological alarm, it sends physiological alarm information according to the time interval;
12. Technical alarm information: When the interface receives a query message as a technical alarm, it sends technical alarm information according to the time interval;
13. Module online and offline information: After establishing a TCP connection between the interface and the client, the interface will send the online module information once, and then send it again if there are any changes afterwards.
14. Parameter alarm configuration information: When the interface receives a query message as a technical alarm, it sends parameter alarm configuration information according to the time interval;

6.2 Broadcast Message(ADT_A01)

The bedside monitor broadcasts an online notification message every second, informing other hosts on the network of its existence. Afterwards, other hosts on the network can dynamically connect based on the broadcast message, and all devices can use the broadcast message.

6.2.1 Format

MSH	Message header
EVN	Event Type
PID	Patient identifier set
PV1	Patient visits
[{ OBX }]	Monitor Name

6.2.2 Example

MSH|^~\&|||||20230523111421||ADT^A01^ADT_A01|1012|P|2.6||||||UTF-8

EVN||20230512094930
PID|||78||Mis^Lucy^Merry||20020202|F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBX||ST|4002^Monitor Name^BHC||Monitor1||||||F

6.2.3 MSH Segment

The MSH segment of the broadcast message has the same definition and usage as the MSH field in section *4.2.2*.

The message type field is ADT_ A01, the message control field is: 1012.

6.2.4 EVN Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
2	24	DTM	R	No	Record Date / Time	yyyymmddhhmmss

Example:EVN||20230512094930

6.2.5 PID Segment and PV1 Segment

The PID and PV1 segments of the broadcast message are identical to the PID and PV1 segments in sections *4.2.3* and *4.2.5*.

6.2.6 OBX Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
2	3	ID	O	No	Value Type	ST
3	705	CWE	R	No	Observation identifier	4002^Monitor Name^BHC Monitor Name Code
5	99999	VARIES	O	Yes	Observations	Monitor Name：Monitor1

Example:OBX||ST|4002^Monitor Name^BHC||Monitor||||||F

6.3 Heartbeat Packet

Both communication parties must send messages to each other every second in the following format. If either party fails to receive a message within 10 seconds, it is considered that a network malfunction has occurred and the network connection is actively interrupted.

6.3.1 Format

MSH Header

Example：
MSH|^~\&|||||20230523112340||ORU^R01^ORU_R01|1014|P|2.6||||||UTF-8

6.3.2 Segment Definition

The message type in the heartbeat packet is: ORU_ R01; Message control ID: 1014

6.4 ACK Message

ACK message is the error message returned by the interface upon receiving an incorrect query message from the client. The client can correct its own query message based on this error message to obtain the desired result message.

6.4.1 Format

MSH Header
MSA Message Acknowledgment

6.4.2 Example

MSH|^~\&|||||20230523120248||ACK|1018|P|2.6||||||UTF-8
MSA|AR|1004|Message format error

6.4.3 MSH Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	1	ST	R	No	Field separator	|
2	4	ST	R	No	Code Characters	^~\&
7	24	DTM	R	No	Date/time of the message	yyyyMMddHHmmss
9	15	MSG	R	No	Message Type	ACK
10	199	ST	R	No	Message Control ID	1018：Control ID of ACK confirmation message
11	3	PT	R	No	Process ID	P

12	60	VID	R	No	Version ID	2.6

MSH|^~\&|||||20230523120248||ACK|1018|P|2.6||||||UTF-8
The control ID of the ACK confirmation message is: ACK

6.4.4 MSA Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	2	ID	R	No	Confirmation Code	AA:Initial mode: Application acceptance; AE:Initial mode: application Software bug; AR:Initial mode:Application Rejection; CA:Application confirmation in enhanced mode, indicating acceptance; CE: Acceptance confirmation in enhanced mode, indicating an error; CR: Acceptance confirmation in enhanced mode, indicating rejection.
2	199	ST	O	No	Message control ID	1004(consistent with the control ID of the query message)
3	80	ST	O	No	Text message	Describe the reason for the error or rejection, such as:Message format error

MSA|AR|1004|Message format error

6.5 Query Message (QRY_R02)

Query messages are messages sent by the server to the interface for querying the required data type.

6.5.1 Format

MSH Message Header
QRD Query Definition
QRF Query filtering

Example:MSH|^~\&|||||20220424095857||ORY^R02^ORY_R02|1029|P|2.6||||||UTF-8
QRD|20220424095857|R|I|2|||||RES
QRF|MON||||0&0^1^0^0

6.5.2 MSH Segment

The MSH segment of the query message has the same definition and usage as the MSH field in section 5.2.3. The message type field of the MSH segment is: QRY_ R02; The message control ID field is the encoding of the data type to be queried by the server, as detailed in *Appendix A.5 Control and Settings.*

Example:

MSH|^~\&|||||20220424095857||ORY^R02^ORY_R02|1029|P|2.6|||||UTF-8

6.5.3 QRD Segment

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	24	DTM	R	No	Query Data/Time	Format:yyyyMMddHHmmss
2	1	ID	R	No	Query	Must always be R
3	1	ID	R	Yes	Query Priority	Must always be I
4	10	ST	R	Yes	Query ID	Cannot be empty and must be less than 16 characters
9	250	CWE	R	No	Filter themes	Must be a result between query and response RES

Example:

QRD|20220424095857|R|I|2|||||RES

Message processing rules:

- The first field must be the correct date/time, otherwise the interface will return an error message;
- The second field must be R, otherwise the interface will return an error message;
- The third field must be I, otherwise the interface will return an error message;
- The fourth field cannot be empty, otherwise the interface will return an error message;
- The ninth field must be RES, otherwise the interface will return an error message.

6.5.4 QRF Segment

QRF defines the detailed content to be queried.

Field	Length	Data Type	Wash ability	Repeat ability	Field Name	Describe
1	20	ST	R	Yes	Topic filtering	Must Always be MON
5	60	ST	O	Yes	Other filtering	Contains locally defined filtering criteria for use between two systems, as detailed in 5.2.5.1

Example:

QRF|MON||||0&0^1^0^0， 1 in the segment indicates that the query information type is physiological parameter data。

Message processing rules:

- The first field must be MON, otherwise the interface will return an error message
- The fifth field cannot be empty, otherwise the interface will return an error message

6.6 Response query report message (ORU_R01)

6.6.1 Periodic parameters

All parameter data is unified by one ORU_ R01 message is sent, and after the query message arrives, it is sent according to the time interval.

6.6.2 MSH Segment

Please refer to *4.2.2* for the definition, where the message type field of the MSH segment is: ORU_ R01; The message control ID field is: 1004.

Example:

MSH|^~\&|||||20220509091226||ORU^R01^ORU_R01|1004|P|2.6||||||UTF-8

6.6.3 PID Segment

The definition and usage of the PID segment here are identical to those in section 4.2.3, please refer to section_ for details.

Example:

PID|||78||Mis^Lucy^Merry||20020202|F

6.6.4 Patient OBX segment

The definition and usage of the PID segment here are identical to those in section 4.2.4, please refer to section *4.2.4* for details.

6.6.5 Format

```
MSH                     Message    Header
PID                     Patient identification
[{
    OBX                 Patient Partial Observations
    ……
}]
PV1                     Patient visits
OBR                     Observation Report ID
{
```

OBX Observations/Results
OBX Observations/Results
.....
}

6.6.6 PV1 Segment

The definition and usage of section PV1 here are identical to section 4.2.5, please refer to section *4.2.5* for details.

Example:

PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A

6.6.7 OBR Segment

The definition and usage of section OBR here are identical to section 4.2.6, please refer to section *4.2.6* for details.

Example:

OBR||||Monitor|||20220510095657

6.6.8 Orderly observation of OBX segments

The definition and usage of section OBR here are identical to section 4.2.7, please refer to section *4.2.7* for details.

Example:

OBX||NM|201^HR^BHC|5001|60||||||F
OBX||NM|202^QTC^BHC|5001|360||||||F
OBX||NM|203^DELTAQTC^BHC|5001|-1||||||F
OBX||NM|204ST-I^BHC|5001|-0.02||||||F
OBX||NM|205ST-II^BHC|5001|-0.02||||||F
OBX||NM|206ST-III^BHC|5001|0.00||||||F
OBX||NM|207ST-aVR^BHC|5001|0.02||||||F
OBX||NM|208ST-aVL^BHC|5001|-0.01||||||F
OBX||NM|209ST-aVF^BHC|5001|-0.01||||||F
OBX||NM|210ST-V1^BHC|5001|0.02||||||F
OBX||NM|211ST-V2^BHC|5001|0.03||||||F
OBX||NM|212ST-V3^BHC|5001|-0.01||||||F
OBX||NM|213ST-V4^BHC|5001|-0.03||||||F

6.6.9 Example

6.6.9.1 Technical alarm

```
MSH|^~\&|||||20220511164311||ORU^R01^ORU_R01|1008|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20220511164311
OBX||CE|0||20307^IBP1 Zero Required^BHC||||||F|||20220511160815
OBX||CE|0||20467^ART Zero Required^BHC||||||F|||20220511160815
OBX||CE|0||23013^No Battery^BHC||||||F|||20220511160823
OBX||CE|0||23008^Recorder Not Exist^BHC||||||F|||20220511160815
```

6.6.9.2 Physiological alarm

```
MSH|^~\&|||||20220512103944||ORU^R01^ORU_R01|1007|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20220512103944
OBX||CE|3||10151^SpO2 High^BHC||||||F|||20220512103917
```

6.6.9.3 Waveform data

```
MSH|^~\&|||||20220512104347||ORU^R01^ORU_R01|1015|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20220512104347
OBX||CD|30021^SPO2^BHC|1|1^^1^^100^0&250||||||F
OBX||TS|30021^SPO2^BHC|1|20220512104347||||||F
OBX||NA|30021^SPO2^BHC|1|208^214^219^221^221^217^213^209^203^197^184^177^17
```

0^164^157^143^136^130^124^118^108^103^99^96^94^92^91^92^93^94^96^97^98^98^97
^95^92^90^86^83^76^73^69^66^63^58^55^52^50^47^43^41^39^37^35^32^31^30^29^28^
26^26^25^25^24^24^24^24^24^24^25^25^25^26^26^26^27^27^28^29^31^32^33^34^35^3
7^38^40^42^45||||||F
OBX||CE|30021^SPO2^BHC|1||||||F

6.6.9.4 Parameter value

MSH|^~\&|||||20220512151316||ORU^R01^ORU_R01|1001|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230607150307
OBX||NM|201^HR^BHC|5001|60||||||F
OBX||NM|220^QT^BHC|5001|356||||||F
OBX||NM|202^QTC^BHC|5001|356||||||F
OBX||NM|203^DELTAQTC^BHC|5001|0||||||F
OBX||NM|204ST-I^BHC|5001|-0.03||||||F
OBX||NM|205ST-II^BHC|5001|-0.02||||||F
OBX||NM|206ST-III^BHC|5001|0.01||||||F
OBX||NM|207ST-aVR^BHC|5001|0.03||||||F
OBX||NM|208ST-aVL^BHC|5001|-0.02||||||F
OBX||NM|209ST-aVF^BHC|5001|-0.01||||||F
OBX||NM|210ST-V1^BHC|5001|0.02||||||F
OBX||NM|211ST-V2^BHC|5001|0.03||||||F
OBX||NM|212ST-V3^BHC|5001|-0.01||||||F
OBX||NM|213ST-V4^BHC|5001|-0.03||||||F
OBX||NM|214ST-V5^BHC|5001|-0.04||||||F
OBX||NM|215ST-V6^BHC|5001|-0.03||||||F
OBX||NM|251^SPO2^BHC|5002|99||||||F
OBX||NM|252^PI^BHC|5002|7.45||||||F
OBX||NM|258^RR^BHC|5002|20||||||F
OBX||NM|259^PR^BHC|5002|60||||||F
OBX||NM|351^NIBP S^BHC|5004|125|11^mmHg^BHC|||||F||APERIODIC|20230607145326
OBX||NM|352^NIBP D^BHC|5004|84|11^mmHg^BHC|||||F||APERIODIC|20230607145326
OBX||NM|353^NIBP M^BHC|5004|96|11^mmHg^BHC|||||F||APERIODIC|20230607145326
OBX||NM|354^NIBP SDP^BHC|5004|41|11^mmHg^BHC|||||F||APERIODIC|20230607145326
OBX||NM|355^NIBP PR^BHC|5004|60|11^mmHg^BHC|||||F||APERIODIC|20230607145326
OBX||NM|401^RR^BHC|5005|20||||||F
OBX||NM|801^CO2Et^BHC|5013|5.0|14^%^BHC|||||F
OBX||NM|802^CO2Fi^BHC|5013|0.2|14^%^BHC|||||F

OBX||NM|803^awRR^BHC|5013|12||||||F
OBX||NM|751^C.O.^BHC|5012|5.2||||||F||APERIODIC|20230607141336
OBX||NM|752^C.I.^BHC|5012|2.9||||||F||APERIODIC|20230607141336
OBX||NM|753^TI^BHC|5012|-1||||||F||APERIODIC|20230607141336
OBX||NM|754^TB^BHC|5012|36.66|21^C^BHC|||||F||APERIODIC|20230607141336
OBX||NM|1501^Sys^BHC|5031|123|11^mmHg^BHC|||||F
OBX||NM|1502^Dia^BHC|5031|82|11^mmHg^BHC|||||F
OBX||NM|1503^Mean^BHC|5031|100|11^mmHg^BHC|||||F
OBX||NM|1504^PR^BHC|5031|60||||||F
OBX||NM|1521^Sys^BHC|5032|123|11^mmHg^BHC|||||F
OBX||NM|1522^Dia^BHC|5032|82|11^mmHg^BHC|||||F
OBX||NM|1523^Mean^BHC|5032|100|11^mmHg^BHC|||||F
OBX||NM|1524^PR^BHC|5032|60||||||F
OBX||NM|1051^T01^BHC|5021|36.6|21^C^BHC|||||F
OBX||NM|1052^T02^BHC|5021|36.5|21^C^BHC|||||F

6.6.9.5 Parameter alarm switch

MSH|^~\&|||||20220520141935||ORU^R01^ORU_R01|1011|P|2.6|||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20220520141935
OBX||NM|4^^BHC|201|1||||||F
OBX||NM|4^^BHC|251|0||||||F
OBX||NM|4^^BHC|351|1||||||F
OBX||NM|4^^BHC|352|0||||||F
OBX||NM|4^^BHC|353|1||||||F
OBX||NM|4^^BHC|354|0||||||F
OBX||NM|4^^BHC|401|1||||||F
OBX||NM|4^^BHC|754|1||||||F
OBX||NM|4^^BHC|1501|1||||||F
OBX||NM|4^^BHC|1502|1||||||F
OBX||NM|4^^BHC|1503|1||||||F
OBX||NM|4^^BHC|1661|1||||||F
OBX||NM|4^^BHC|1662|0||||||F
OBX||NM|4^^BHC|1663|0||||||F
OBX||NM|4^^BHC|1051|1||||||F
OBX||NM|4^^BHC|1052|1||||||F

6.6.9.6 Parameter alarm level

```
MSH|^~\&|||||20220520142146||ORU^R01^ORU_R01|1010|P|2.6||||||UTF-8
PID|||78||Mis^Lucy^Merry||20020202|F
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20220520142146
OBX||NM|3^^BHC|201|3||||||F
OBX||NM|3^^BHC|251|3||||||F
OBX||NM|3^^BHC|351|3||||||F
OBX||NM|3^^BHC|352|2||||||F
OBX||NM|3^^BHC|353|3||||||F
OBX||NM|3^^BHC|354|2||||||F
OBX||NM|3^^BHC|401|2||||||F
OBX||NM|3^^BHC|754|1||||||F
OBX||NM|3^^BHC|1501|2||||||F
OBX||NM|3^^BHC|1502|3||||||F
OBX||NM|3^^BHC|1503|2||||||F
OBX||NM|3^^BHC|1661|2||||||F
OBX||NM|3^^BHC|1662|2||||||F
OBX||NM|3^^BHC|1663|2||||||F
OBX||NM|3^^BHC|1051|1||||||F
OBX||NM|3^^BHC|1052|1||||||F
```

6.6.9.7 Parameter alarm level

```
MSH|^~\&|||||20230713104818||ORU^R01^ORU_R01|1009|P|2.6||||||UTF-8
PID||||||||M
OBX||NM|4201^Height^BHC||66.9|32^inch^BHC|||||F
OBX||NM|4202^Weight^BHC||143.3|42^lb^BHC|||||F
OBX||NM|4203^Blood^BHC||4||||||F
OBX||NM|4204^Pace^BHC||2||||||F
PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A
OBR||||Monitor|||20230713104818
OBX||NM|1^^BHC|201|120||||||F
OBX||NM|2^^BHC|201|50||||||F
OBX||NM|1^^BHC|251|100||||||F
OBX||NM|2^^BHC|251|90||||||F
OBX||NM|1^^BHC|351|160|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|351|90|11^mmHg^BHC|||||F
```

OBX||NM|1^^BHC|352|100|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|352|60|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|353|115|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|353|70|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|354|60|11^mmHg^BHC|||||F
OBX||NM|2^^BHC|354|20|11^mmHg^BHC|||||F
OBX||NM|1^^BHC|401|30||||||F
OBX||NM|2^^BHC|401|8||||||F
OBX||NM|1^^BHC|801|6.6|14^%^BHC|||||F
OBX||NM|2^^BHC|801|3.3|14^%^BHC|||||F
OBX||NM|1^^BHC|802|0.5|14^%^BHC|||||F
OBX||NM|2^^BHC|802|0.5|14^%^BHC|||||F
OBX||NM|1^^BHC|803|30||||||F
OBX||NM|2^^BHC|803|8||||||F
OBX||NM|1^^BHC|754|39.0|21^C^BHC|||||F
OBX||NM|2^^BHC|754|36.0|21^C^BHC|||||F
OBX||NM|1^^BHC|1501|218|13^cmH2O^BHC|||||F
OBX||NM|2^^BHC|1501|122|13^cmH2O^BHC|||||F
OBX||NM|1^^BHC|1502|122|13^cmH2O^BHC|||||F
OBX||NM|2^^BHC|1502|68|13^cmH2O^BHC|||||F
OBX||NM|1^^BHC|1503|150|13^cmH2O^BHC|||||F
OBX||NM|2^^BHC|1503|95|13^cmH2O^BHC|||||F
OBX||NM|1^^BHC|1521|21.3|12^kPa^BHC|||||F
OBX||NM|2^^BHC|1521|12.0|12^kPa^BHC|||||F
OBX||NM|1^^BHC|1522|12.0|12^kPa^BHC|||||F
OBX||NM|2^^BHC|1522|6.7|12^kPa^BHC|||||F
OBX||NM|1^^BHC|1523|14.7|12^kPa^BHC|||||F
OBX||NM|2^^BHC|1523|9.3|12^kPa^BHC|||||F
OBX||NM|1^^BHC|1051|39.0|21^C^BHC|||||F
OBX||NM|2^^BHC|1051|36.0|21^C^BHC|||||F

7 ADT Net Query Interface

This section introduces the content of the ADT query interface, which mainly involves the communication process between the ADT query interface and the BltADTServer server of Biolight development.

7.1 Communication Process

7.1.1 Network Connection

This interface is the client that establishes the TCP connection, and BltADTServer is the server that listens.

After a TCP connection is set, the TCP interface does not actively close the connection. If the TCP connection is interrupted or closed, the interface automatically reconnects to the TCP within a fixed period of time.

All ADT messages from both sides are transmitted in the same TCP connection. The following two sections detail the data communication process in both transmission directions.

7.1.2 Data Interaction

This interface sends a query message in QRY^A19 format. The QRY^A19 message contains a patient medical record number (MRN) as a unique query condition.

The BltADTServer server executes the exact query and returns an ADR^A19 message based on the query criteria. If patient information is found that matches the query criteria, patient records that meet the query criteria are returned. If no patient information is found that matches the query criteria, the message returned does not contain any patient information. If the HIS/CIS system does not return any message within a limited period of time, the interface treats it as a query timeout.

After receipt of the HL7 message, the interface returns an ACK message indicating whether the format is correct. This interface does not perform any operation when receiving an ACK message.

7.2 ADT Query Message (QRY_A19)

7.2.1 Format

```
MSH        Message Header
QRD        Query Definition
```

7.2.2 Example

MSH|^~\\&||||ADTServer|20220906095605||QRY^A19^QRY_A19|1016|P|2.6||||||UTF-8
QRD|20220906095605|R|I|2|||1RD|M2021|^DEM|^BLTCMS

Among them, the information of ADT patient whose medical record number is M2021 is known to be queried in the QRD_8 field.

7.2.3 MSH Segment

For details, see *4.2.2* when the message type is QRY_A19 and the message control ID field is 1016. For details, see Appendix *A.5 Controls and Settings*.

7.2.4 QRD Segment

SEQ	LEN	DT	OPT	REP	Field Name	Description
1	24	DTM	R	No	Query the date/time	Format：yyyyMMddHHmmss
2	1	ID	R	No	Query mode	Must be R
3	1	ID	R	Yes	Query priority	Must be I
4	10	ST	R	Yes	Query ID	The value cannot be empty and must be less than 16 characters long
7	10	CQ	R	Yes	Quantity request limit	Can't be empty,1RD
8	60	XCN	R	Yes	Who to Filter	Patient unique identification
9	250	CWE	R	No	Filter topic	The value cannot be empty and represents the request data type. DEM: Demographic data
10			R	No	Department data code	“BLTCMS”

7.3 ADT Query Reply Message (ADR_A19)

7.3.1 Format

- Find the patient

```
MSH                 Message  Header
MSA                 Message  Acknowledgment
QRD                 Query  definition
[
    PID             Patient  Identification
    PV1             Patient  Visit
    {
        OBX         Observation/Result
    }
]
```

- No patient found

```
MSH       Message  Header
MSA       Message  Acknowledgment
QRD       Query  definition
```

7.3.2 Example

- Find the patient

MSH|^~\\&||||20220906111243||ADR^A19^ADR_A19|1017|P|2.6|||||UTF-8
MSA|AA|1016|Query Found
QRD|20220906095605|R|I|2|||1RD|M2021|^DEM|^BLTCMS
PID|||M2021||Mis^Luck^Bob||19951108|F
PV1||I|CCU^R1^B4&36&0||||DoctorInfo||||||||||U
OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F
OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F
OBX||NM|4203^Blood^BHC||3||||||F
OBX||NM|4204^Pace^BHC||2||||||F

- No patient found

MSH|^~\\&||||20220906110802||ADR^A19^ADR_A19|1017|P|2.6|||||UTF-8
MSA|AA|1016|No Query Found
QRD|20220906095605|R|I|2|||1RD|M2021|^DEM|^BLTCMS

7.3.3 MSH Segment

For details, see *4.2.2* when the message type is ADR_A19 and the message control ID field is 1017. For details, see Appendix *A.5 Controls and Settings*.

7.3.4 MSA Segment

SEQ	LEN	DT	OPT	REP	Field Name	Description
1	2	ID	R	Yes	Confirmation code	AA： Application acceptance (affirmative confirmation) AR： Application acceptance (negative confirmation) If the QRY^A19 message format is correct, this field is "AA", otherwise "AR"
2	20	ST	R	Yes	Message control ID	The message control ID accompanying the QRY query message containing the request. This allows the client to map the response message to the corresponding request.
3	80	ST	O	Yes	Text message	If the queried patient is Found, the typical text will be "Query Found." If the queried patient is not Found, the typical text will be "No Query Found."

7.3.5 QRD Segment

The format and content are the same as those received by QRY^A19.

7.3.6 PID Segment

The definition and usage of the PID segment here are exactly the same as in Section 4.2.3, see Section *4.2.3* for details.

7.3.7 PV1 Segment

SEQ	LEN	DT	OPT	REP	Field Name	Description
2	1	IS	R	No	I:Inpatient O:Outpatient	I:Inpatient O:Outpatient
3	80	PL	O	No	<point of care> ^<room>^<bed> <bed> divided into: <OfficeName>&<bed id>&<IP>&<IPSeq>&0 <IP>:the bedside device	<point of care> ^<room>^<bed> <bed> divided into: <OfficeName>&<bed id>&<IP>&<IPSeq>&0 <IP>:the bedside device IP

					IP Address; <IPSeq>:the telemetry transmitter's sequence number.	Address; <IPSeq>:the telemetry transmitter's sequence number.
7	250	XCN	O	Yes	Attending doctor	
18	2	IS	O	Y	Patient type	Neonate：N Adult：A Ped：P Unknown：U
44	24	DTM	O	Y	Admission date	Format:yyyyMMddHHmmss

7.3.8 OBX Segment

The demographic data of some patients were in PID segment and PV1 segment. Other measurements were performed using the OBX segment, including patient height, patient weight, blood type, and pacing markers.

The definition and usage of OBX here are the same as in Section 4.2.7, see *4.2.7* for details.

7.4 ACK Message

7.4.1 Format

```
MSH     Message  Header
MSA     Message  Acknowledgment
```

7.4.2 Example

MSH|^~\\&||||20220906115151||ACK|1018|P|2.6||||||UTF-8
MSA|AA|1016

7.4.3 MSH Segment

For details, see *4.2.2*. The Message control ID field is 1018. For details, see Appendix *A.5 Controls and Settings*.

7.4.4 MSA Segment

The definition and usage of OBX here are the same as in Section 7.3.4, see *7.3.4* for details.

Appendix A Biolight Coding System

Appendix A.1 Parameter ID

ID	Content	Module	Description	Default unit
201	HR	ECG	Heart rate	bpm
202	QTc	ECG	QT parameter: Corrected QT interval	mv
203	Delta QTc	ECG	QT parameter	mv
204	ST-I	ECG	The ST segment is the period from the end of ventricular depolarization to the beginning of ventricular repolarization, or the end of the QRS wave group (J-point) to the beginning of the T wave. ST parameters corresponding to different ECG lead types are as follows: (1) 3 Lead：ST-I、ST-II、ST-III、ST-AVR 、 ST-AVL 、 ST-AVF; (2) 5 Lead：ST-I、ST-II、ST-III、ST-AVR 、 ST-AVL 、 ST-AVF、ST-V; (3) 6 Lead：ST-I、ST-II、ST-III、ST-AVR 、 ST-AVL 、 ST-AVF、ST-Va、ST-Vb; (4) 12 Lead：ST-I、ST-II、ST-III、ST-AVR 、 ST-AVL 、 ST-AVF、ST-V1、ST-V2、ST-V3 、 ST-V4 、 ST-V5 、 ST-V6；	mv
205	ST-II	ECG		mv
206	ST-III	ECG		mv
207	ST-AVR	ECG		mv
208	ST-AVL	ECG		mv
209	ST-AVF	ECG		mv
210	ST-V1	ECG		mv
211	ST-V2	ECG		mv
212	ST-V3	ECG		mv
213	ST-V4	ECG		mv
214	ST-V5	ECG		mv
215	ST-V6	ECG		mv
216	ST-V	ECG		mv
217	ST-Va	ECG		mv
218	ST-Vb	ECG		mv
219	PVcs	ECG	Number of ventricular contractions	/min
220	QT	ECG	QT parameter	mv
251	SPO2	SPO2	Percent Oxyhemoglobin Saturation - Arterial	%
252	PI	SPO2	Perfusion Index	
253	PVI	SPO2	It's the variation of PI over a breathing cycle	
254	SPCO	SPO2	carboxyhemoglobin	

255	SPOC	SPO2	Oxygen content	
256	SPHB	SPO2	Non-invasive hemoglobin	
257	SPMET	SPO2	methemoglobin	
258	RR	SPO2	Respiratory rate	rmp
259	PR	SPO2	Pulse Rate	bpm
301	SPO2	SPO2L	Percent Oxyhemoglobin Saturation - Arterial	%
302	PI	SPO2L	Perfusion Index	
303	PVI	SPO2L	It's the variation of PI over a breathing cycle	
304	SPCO	SPO2L	carboxyhemoglobin	
305	SPOC	SPO2L	Oxygen content	
306	SPHB	SPO2L	Non-invasive hemoglobin	
307	SPMET	SPO2L	methemoglobin	
308	RR	SPO2L	Respiratory rate	rmp
309	PR	SPO2L	Pulse Rate	bpm
310	Delta SPO2	SPO2L	Pulse oxygen saturation difference	
351	SYS	NIBP	Noninvasive blood pressure systolic blood pressure	mmHg
352	DIA	NIBP	Noninvasive blood pressure diastolic blood pressure	mmHg
353	MAP	NIBP	Mean arterial pressure of noninvasive blood pressure	mmHg
354	SDP	NIBP	Pulse pressure difference	mmHg
355	PR	NIBP	Pulse rate	mmHg
401	RR	RESP	Respiratory rate	rmp
451	C.O.	ICG	Cardiac output	L/min
452	C.I.	ICG	Cardiac Index	L/min/m²
453	SV	ICG	Stroke volume	ml
454	SI	ICG	Stroke index	ml//m²
455	SVR	ICG	Systemic vascular resistance	dyn*s/cm^5
456	SVRI	ICG	Systemic vascular resistance index	dyn*s*m^2/cm^5
457	TFC	ICG	Thoracic fluid content	/kΩ
458	TFI	ICG	Thoracic fluid index	Ω
459	HR	ICG	Heart rate	bpm
460	QI	ICG	Quality index	%
461	DO2I	ICG	Oxygen delivery index	ml/min/m^2
501	BIS	BIS	Bispectral Index	
502	EMG	BIS	Electromyograph	dB
503	SQI	BIS	Signal quality index	%

504	SR	BIS	Rejection ratio	%
505	SEF	BIS	Spectral edge frequency	Hz
506	TP	BIS	Total power	dB
507	BC	BIS	Burst count	bpm
551	SEF	EEG1	Spectral edge frequency	Hz
552	MDF	EEG1	Median frequency	Hz
553	PPF	EEG1	Peak power frequency	Hz
554	TP	EEG1	Total power	dB
~~555~~	~~AMP~~	~~EEG1~~		
556	Delta	EEG1	Delta wave relative power	%
557	Theta	EEG1	Theta wave relative power	%
558	Alpha	EEG1	Alpha wave relative power	%
559	Beta	EEG1	Beta wave relative power	%
571	SEF	EEG2	Spectral edge frequency	Hz
572	MDF	EEG2	Median frequency	Hz
573	PPF	EEG2	Peak power frequency	Hz
574	TP	EEG2	Total power	dB
~~575~~	~~AMP~~	~~EEG2~~		
576	Delta	EEG2	Delta wave relative power	%
577	Theta	EEG2	Theta wave relative power	%
578	Alpha	EEG2	Alpha wave relative power	%
579	Beta	EEG2	Beta wave relative power	%
591	SEF	EEG3	Spectral edge frequency	Hz
592	MDF	EEG3	Median frequency	Hz
593	PPF	EEG3	Peak power frequency	Hz
594	TP	EEG3	Total power	dB
~~595~~	~~AMP~~	~~EEG3~~		
596	Delta	EEG3	Delta wave relative power	%
597	Theta	EEG3	Theta wave relative power	%
598	Alpha	EEG3	Alpha wave relative power	%
599	Beta	EEG3	Beta wave relative power	%
611	SEF	EEG4	Spectral edge frequency	Hz
612	MDF	EEG4	Median frequency	Hz
613	PPF	EEG4	Peak power frequency	Hz
614	TP	EEG4	Total power	dB
615	~~AMP~~	~~EEG4~~		
616	Delta	EEG4	Delta wave relative power	%
617	Theta	EEG4	Theta wave relative power	%
618	Alpha	EEG4	Alpha wave relative power	%
619	Beta	EEG4	Beta wave relative power	%
751	C.O.	C.O.	Cardiac output	L/min
752	C.I.	C.O.	Cardiac index	L/min/m²

753	TI	C.O.	Injection temperature	℃
754	TB	C.O.	Blood Temperature	℃
801	CO2ET	CO2	End-tidal carbon dioxide	mmHg
802	CO2FI	CO2	Fraction of inspired carbon	mmHg
803	AwRR	CO2	Airway respiratory rate	rpm
821	N2OET	N2O	End-of-breath nitrous oxide	%
822	N2OFI	N2O	Fraction of inspired nitrous oxide	%
841	O2ET	O2	End-tidal oxygen	mmHg
842	O2FI	O2	Fraction of inspired oxygen	mmHg
861	AAET	AA	End-tidalAnesthetic gas	%
862	AAFI	AA	Fraction of inspired Anesthetic gas	%
871	HALET	AA	End-of-breath anesthetic gas halothane	%
872	HALFI	AA	Inhalation of the anesthetic gas halothane	%
881	ENFET	AA	The anesthetic gas enflurane at the end of breath	%
882	ENFFI	AA	Inhalation of the anesthetic gas enflurane	%
891	ISOET	AA	Isoflurane, the anesthetic gas at the end of breath	%
892	ISOFI	AA	Inhalation of the anesthetic gas isoflurane	%
901	SEVET	AA	The anesthetic gas sevoflurane at the end of breath	%
902	SEVFI	AA	Inhalation of the anesthetic gas sevoflurane	%
911	DESET	AA	The anesthetic gas desflurane at the end of breath	%
912	DESFI	AA	Inhalation of the anesthetic gas desflurane	%
921	MAC	AA	Minimum alveolar effective concentration	/
1001	Tvi	RM	Inspiratory tidal volume	ml
1002	Tve	RM	Expiratory tidal volume	ml
1003	Mvi	RM	Inhalation ventilation per minute	L/min
1004	Mve	RM	Exhaled volume per minute	L/min
1005	PIP	RM	Peak inspiratory pressure	cmH_2O
1006	Pplat	RM	Platform pressure value	cmH_2O
1007	PEEP	RM	Positive end expiratory pressur	cmH_2O

1008	Pmean	RM	Mean airway pressure	cmH_2O
1009	Compl	RM	Compliance	ml/cmH_2O
1010	I:E	RM	Inspiatory time: expiratory time ratio	/
1011	RR	RM	Respiration rate	rpm
1012	Resi	RM	Airway resistance	$cmH_2O/L/s$
1013	TV1	RM	First second expiratory rate	%
1051	T01	TEMP_A	Temperature of channel 1	℃
1052	T02	TEMP_A	Temperature of channel 2	℃
1053	T03	TEMP_A	Temperature of channel 3	℃
1054	T04	TEMP_A	Temperature of channel 4	℃
1055	T05	TEMP_A	Temperature of channel 5	℃
1056	T06	TEMP_A	Temperature of channel 6	℃
1057	T07	TEMP_A	Temperature of channel 7	℃
1058	T08	TEMP_A	Temperature of channel 8	℃
1059	T09	TEMP_A	Temperature of channel 9	℃
1060	T010	TEMP_A	Temperature of channel 10	℃
1061	TSkin	TEMP_A	Skin temperature	℃
1062	TCore	TEMP_A	Core temperature	℃
1063	TAxil	TEMP_A	Axillary temperature	℃
1064	TNaso	TEMP_A	Nasopharyngeal temperature	℃
1065	TEso	TEMP_A	Esophagus temperature	℃
1066	TRect	TEMP_A	Rectal temperature	℃
1067	TAmb	TEMP_A	Ambient temperature	℃
1068	TAirw	TEMP_A	Airway temperature	℃

1069	TVesic	TEMP_A	vesica urinaria temperature	℃
1070	TBlood	TEMP_A	Temperature of blood	℃
1071	TMyo	TEMP_A	Myocardial temperature	℃
1072	TTymp	TEMP_A	Tympanic membrane temperature	℃
1073	TCereb	TEMP_A	Cerebrum temperature	℃
1101	T01	TEMP_B		
1102	T02	TEMP_B		
1103	T03	TEMP_B		
1104	T04	TEMP_B		
1105	T05	TEMP_B		
1106	T06	TEMP_B		
1107	T07	TEMP_B		
1108	T08	TEMP_B		
1109	T09	TEMP_B		
1110	T010	TEMP_B		
1111	TSkin	TEMP_B		
1112	TCore	TEMP_B		
1113	TAxil	TEMP_B		
1114	TNaso	TEMP_B		
1115	TEso	TEMP_B		
1116	TRect	TEMP_B		
1117	TAmb	TEMP_B		

1118	TAirw	TEMP_B		
1119	TVesic	TEMP_B		
1120	TBlood	TEMP_B		
1121	TMyo	TEMP_B		
1122	TTymp	TEMP_B		
1123	TCereb	TEMP_B		
1151	T01	TEMP_C		
1152	T02	TEMP_C		
1153	T03	TEMP_C		
1154	T04	TEMP_C		
1155	T05	TEMP_C		
1156	T06	TEMP_C		
1157	T07	TEMP_C		
1158	T08	TEMP_C		
1159	T09	TEMP_C		
1160	T010	TEMP_C		
1161	TSkin	TEMP_C		
1162	TCore	TEMP_C		
1163	TAxil	TEMP_C		
1164	TNaso	TEMP_C		
1165	TEso	TEMP_C		
1166	TRect	TEMP_C		

1167	TAmb	TEMP_C		
1168	TAirw	TEMP_C		
1169	TVesic	TEMP_C		
1170	TBlood	TEMP_C		
1171	TMyo	TEMP_C		
1172	TTymp	TEMP_C		
1173	TCereb	TEMP_C		
1201	T01	TEMP_D		
1202	T02	TEMP_D		
1203	T03	TEMP_D		
1204	T04	TEMP_D		
1205	T05	TEMP_D		
1206	T06	TEMP_D		
1207	T07	TEMP_D		
1208	T08	TEMP_D		
1209	T09	TEMP_D		
1210	T010	TEMP_D		
1211	TSkin	TEMP_D		
1212	TCore	TEMP_D		
1213	TAxil	TEMP_D		
1214	TNaso	TEMP_D		
1215	TEso	TEMP_D		

1216	TRect	TEMP_D		
1217	TAmb	TEMP_D		
1218	TAirw	TEMP_D		
1219	TVesic	TEMP_D		
1220	TBlood	TEMP_D		
1221	TMyo	TEMP_D		
1222	TTymp	TEMP_D		
1223	TCereb	TEMP_D		
1251	T01	TEMP_E		
1252	T02	TEMP_E		
1253	T03	TEMP_E		
1254	T04	TEMP_E		
1255	T05	TEMP_E		
1256	T06	TEMP_E		
1257	T07	TEMP_E		
1258	T08	TEMP_E		
1259	T09	TEMP_E		
1260	T010	TEMP_E		
1261	TSkin	TEMP_E		
1262	TCore	TEMP_E		
1263	TAxil	TEMP_E		
1264	TNaso	TEMP_E		

1265	TEso	TEMP_E		
1266	TRect	TEMP_E		
1267	TAmb	TEMP_E		
1268	TAirw	TEMP_E		
1269	TVesic	TEMP_E		
1270	TBlood	TEMP_E		
1271	TMyo	TEMP_E		
1272	TTymp	TEMP_E		
1273	TCereb	TEMP_E		
1501	Sys	IBP_P1	IBP P1 Module systolic blood pressure	mmHg
1502	Dia	IBP_P1	IBP P1 Module diastolic blood pressure	mmHg
1503	Mean	IBP_P1	IBP P1 Module mean pressure	mmHg
1504	PR	IBP_P1	IBP P1 Module pulse rate	bpm
1505	PPV	IBP_P1	IBP P1 Module Pulse pressure variation rate	%
1521	Sys	IBP_P2	IBP P2 Module systolic blood pressure	mmHg
1522	Dia	IBP_P2	IBP P2 Module diastolic blood pressure	mmHg
1523	Mean	IBP_P2	IBP P2 Module mean pressure	mmHg
1524	PR	IBP_P2	IBP P2 Module pulse rate	bpm
1525	PPV	IBP_P2	IBP P2 Module Pulse pressure variation rate	%
1541	Sys	IBP_P3	IBP P3 Module systolic blood pressure	mmHg
1542	Dia	IBP_P3	IBP P3 Module diastolic blood pressure	mmHg
1543	Mean	IBP_P3	IBP P3 Module mean pressure	mmHg
1544	PR	IBP_P3	IBP P3 Module pulse rate	bpm
1545	PPV	IBP_P3	IBP P3 Module Pulse pressure variation rate	%
1561	Sys	IBP_P4	IBP P4 Module systolic blood	mmHg

			pressure	
1562	Dia	IBP_P4	IBP P4 Module diastolic blood pressure	mmHg
1563	Mean	IBP_P4	IBP P4 Module mean pressure	mmHg
1564	PR	IBP_P4	IBP P4 Module pulse rate	bpm
1565	PPV	IBP_P4	IBP P4 Module Pulse pressure variation rate	%
1581	Sys	IBP_P5	IBP P5 Module systolic blood pressure	mmHg
1582	Dia	IBP_P5	IBP P5 Module diastolic blood pressure	mmHg
1583	Mean	IBP_P5	IBP P5 Module mean pressure	mmHg
1584	PR	IBP_P5	IBP P5 Module pulse rate	bpm
1585	PPV	IBP_P5	IBP P5 Module Pulse pressure variation rate	%
1601	Sys	IBP_P6	IBP P6 Module systolic blood pressure	mmHg
1602	Dia	IBP_P6	IBP P6 Module diastolic blood pressure	mmHg
1603	Mean	IBP_P6	IBP P6 Module mean pressure	mmHg
1604	PR	IBP_P6	IBP P6 Module pulse rate	bpm
1605	PPV	IBP_P6	IBP P6 Module Pulse pressure variation rate	%
1621	Sys	IBP_P7	IBP P7 Module systolic blood pressure	mmHg
1622	Dia	IBP_P7	IBP P7 Module diastolic blood pressure	mmHg
1623	Mean	IBP_P7	IBP P7 Module mean pressure	mmHg
1624	PR	IBP_P7	IBP P7 Module pulse rate	bpm
1625	PPV	IBP_P7	IBP P7 Module Pulse pressure variation rate	%
1641	Sys	IBP_P8	IBP P8 Module systolic blood pressure	mmHg
1642	Dia	IBP_P8	IBP P8 Module diastolic blood pressure	mmHg
1643	Mean	IBP_P8	IBP P8 Module mean pressure	mmHg
1644	PR	IBP_P8	IBP P8 Module pulse rate	bpm
1645	PPV	IBP_P8	IBP P8 Module Pulse pressure variation rate	%
1661	Sys	IBP_ART	IBP ART Module systolic blood pressure	mmHg
1662	Dia	IBP_AR	IBP ART Module diastolic blood	mmHg

		T	pressure	
1663	Mean	IBP_ART	IBP ART Module mean pressure	mmHg
1664	PR	IBP_ART	IBP ART Module pulse rate	bpm
1665	PPV	IBP_ART	IBP ART Module Pulse pressure variation rate	%
1681	Sys	IBP_CVP	IBP CVP Module systolic blood pressure	mmHg
1682	Dia	IBP_CVP	IBP CVP Module diastolic blood pressure	mmHg
1683	Mean	IBP_CVP	IBP CVP Module mean pressure	mmHg
1684	PR	IBP_CVP	IBP CVP Module pulse rate	bpm
1685	PPV	IBP_CVP	IBP CVP Module Pulse pressure variation rate	%
1701	Sys	IBP_PA	IBP PA Module systolic blood pressure	mmHg
1702	Dia	IBP_PA	IBP PA Module diastolic blood pressure	mmHg
1703	Mean	IBP_PA	IBP PA Module mean pressure	mmHg
1704	PR	IBP_PA	IBP PA Module pulse rate	bpm
1705	PPV	IBP_PA	IBP PA Module Pulse pressure variation rate	%
1706	Pawp	IBP_PA	IBP PA Module Pulmonary artery wedge pressure	mmHg
1721	Sys	IBP_RAP	IBP RAP Module systolic blood pressure	mmHg
1722	Dia	IBP_RAP	IBP RAP Module diastolic blood pressure	mmHg
1723	Mean	IBP_RAP	IBP RAP Module mean pressure	mmHg
1724	PR	IBP_RAP	IBP RAP Module pulse rate	bpm
1725	PPV	IBP_RAP	IBP RAP Module Pulse pressure variation rate	%
1741	Sys	IBP_LAP	IBP LAP Module systolic blood pressure	mmHg
1742	Dia	IBP_LAP	IBP LAP Module diastolic blood pressure	mmHg
1743	Mean	IBP_LAP	IBP LAP Module mean pressure	mmHg

1744	PR	IBP_LAP	IBP LAP Module pulse rate	bpm
1745	PPV	IBP_LAP	IBP LAP Module Pulse pressure variation rate	%
1761	Sys	IBP_ICP	IBP ICP Module systolic blood pressure	mmHg
1762	Dia	IBP_ICP	IBP ICP Module diastolic blood pressure	mmHg
1763	Mean	IBP_ICP	IBP ICP Module mean pressure	mmHg
1764	PR	IBP_ICP	IBP ICP Module pulse rate	bpm
1765	PPV	IBP_ICP	IBP ICP Module Pulse pressure variation rate	%
1781	Sys	IBP_UAP	IBP UAP Module systolic blood pressure	mmHg
1782	Dia	IBP_UAP	IBP UAP Module diastolic blood pressure	mmHg
1783	Mean	IBP_UAP	IBP UAP Module mean pressure	mmHg
1784	PR	IBP_UAP	IBP UAP Module pulse rate	bpm
1785	PPV	IBP_UAP	IBP UAP Module Pulse pressure variation rate	%
1801	Sys	IBP_BAP	IBP BAP Module systolic blood pressure	mmHg
1802	Dia	IBP_BAP	IBP BAP Module diastolic blood pressure	mmHg
1803	Mean	IBP_BAP	IBP BAP Module mean pressure	mmHg
1804	PR	IBP_BAP	IBP BAP Module pulse rate	bpm
1805	PPV	IBP_BAP	IBP BAP Module Pulse pressure variation rate	%
1821	Sys	IBP_FAP	IBP FAP Module systolic blood pressure	mmHg
1822	Dia	IBP_FAP	IBP FAP Module diastolic blood pressure	mmHg
1823	Mean	IBP_FAP	IBP FAP Module mean pressure	mmHg
1824	PR	IBP_FAP	IBP FAP Module pulse rate	bpm
1825	PPV	IBP_FAP	IBP FAP Module Pulse pressure variation rate	%
1841	Sys	IBP_UV	IBP UVP Module systolic blood	mmHg

		P	pressure	
1842	Dia	IBP_UVP	IBP UVP Module diastolic blood pressure	mmHg
1843	Mean	IBP_UVP	IBP UVP Module mean pressure	mmHg
1844	PR	IBP_UVP	IBP UVP Module pulse rate	bpm
1845	PPV	IBP_UVP	IBP UVP Module Pulse pressure variation rate	%
1861	Sys	IBP_LA	IBP LA Module systolic blood pressure	mmHg
1862	Dia	IBP_LA	IBP LA Module diastolic blood pressure	mmHg
1863	Mean	IBP_LA	IBP LA Module mean pressure	mmHg
1864	PR	IBP_LA	IBP LA Module pulse rate	bpm
1865	PPV	IBP_LA	IBP LA Module Pulse pressure variation rate	%
1881	Sys	IBP_AO	IBP AO Module systolic blood pressure	mmHg
1882	Dia	IBP_AO	IBP AO Module diastolic blood pressure	mmHg
1883	Mean	IBP_AO	IBP AO Module mean pressure	mmHg
1884	PR	IBP_AO	IBP AO Module pulse rate	bpm
1885	PPV	IBP_AO	IBP AO Module Pulse pressure variation rate	%
2001	rSO2	RSO2_1	Cerebral and regional oxygen saturation	%
2002	BL	RSO2_1	Baseline	
2003	DelteBL	RSO2_1	Percent change from baseline in rSO2	%
2004	AUC	RSO2_1	Area under the curve	%*min
2005	DelteSpO2	RSO2_1	Blood oxygen saturation and cerebral oxygen difference	%
2006	AVG	RSO2_1	rSO2 averaged over the last 60 minutes	
2007	Position	RSO2_1	Number of channels	
2021	rSO2_2	RSO2_2	Cerebral and regional oxygen saturation	%
2022	BL_2	RSO2_2	Baseline	
2023	DelteBL_2	RSO2_2	Percent change from baseline in rSO2	%
2024	AUC_2	RSO2_2	Area under the curve	%*min

2025	DelteSpO2_2	RSO2_2	Blood oxygen saturation and cerebral oxygen difference	%
2026	AVG_2	RSO2_2	rSO2 averaged over the last 60 minutes	
2027	Position_2	RSO2_2	Number of channels	
2051	Total Score	EWS	Total score	
2052	Score Type	EWS	Score type	
2101	Total Score	GCS	Total score	
2102	Eye Opening	GCS	Single item score of eye-opening response	
2103	Verbal Response	GCS	Single item scores for verbal response	
2104	Motor Response	GCS	Single item scores for motor response	
2151	Drop Speed	DM	Drop speed	Drops/min
2152	Flowing	DM	Flow velocity	ml/h
2201	Stimulate Current	NMT	Stimulate current	mA
2202	Measure Current	NMT	Measure current	mA
2203	Bargraph Strength	NMT	Bargraph strength	
2204	Measure Result	NMT	Measure result	
2205	Result Type	NMT	Result type（0：Score； 1：%）	
2206	Stimulate Cycle	NMT	Stimulate cycle	
2207	Stimulate Mode	NMT	Stimulate mode	
2208	Auto Mode	NMT	The depth of block in Auto mode is indicated	
2209	Bargraph Number	NMT	Bargraph number	
4001	Standby Mode	SYSTEM		
4002	Monitor Name	SYSTEM		
4003	Central monitoring system Name	SYSTEM		
4004	Central monitoring system system time	SYSTEM		
4005	Sn	SYSTEM		
4006	It bed connection status	SYSTEM		
4021	Highest level of technical alarm	SYSTEM		
4022	Highest level of	SYSTE		

	physiological alarm	M		
4023	System alarm status	SYSTEM		
4024	ECG Lead type	SYSTEM		
4025	PR Source	SYSTEM		
4026	RR Source	SYSTEM		

Appendix A.2 Parameter attribute section

ID	Sub ID	Content	Describe	Value
1	Parameter ID	Alarm out of limit	Alarm Upper Limit Specify with child ID Parameter ID	
2	Parameter ID	Alarm below limit	Alarm Lower Limit	
3	Parameter ID	Alarm level	Alarm Level	0:Tips 1:Low level alarm 2:Middle alarm 3:High alarm
4	Parameter ID	Alarm switch	Alarm Switch	
5	Parameter ID	Online module	Module online event（Module Lode）: Example:OBX||CE|5^^BHC||5002^SPO2^BHC||||||F SpO2 Module is online	
6	Parameter ID	Offline module	Module offline event（Module Unlode）: Example: OBX||CE|6^^BHC||5002^SPO2^BHC||||||F SPO2 Module offline	
7	Parameter ID	Supported parameters	Parameters supported by the module（Parameter Lode） Example: OBX||CE|7^^BHC|5002|259^PR^BHC||||||F	

			SPO2 The module supports PR parameters	
8	Parameter ID	Unsupported parameters	Parameter not supported by module（Parameter Unlode） Example: OBX||CE|8^^BHC|5002|259^PR^BHC||||||F SPO2 The module does not support PR parameters	

Appendix A.3 Module ID section

ID	Content	Description
5001	ECG	ECG (Electrocardiogram)module ID
5002	SPO2	SPO2(Pulse oxygen saturation) module ID
5003	SPO2L	SPO2L(Pulse oxygen saturation) module ID
5004	NIBP	NIBP(Non-invasive Blood pressure) module ID
5005	RESP	RESP(Respiratory) module ID
5006	ICG	ICG(Invasive cardiac output) module ID
5007	BIS	BIS(Bispectral index) module ID
5008	EEG1	EEG (Electroencephalogram) channel 1 module ID
5009	EEG2	EEG (Electroencephalogram) channel 2 module ID
5010	EEG3	EEG (Electroencephalogram) channel 3 module ID
5011	EEG4	EEG (Electroencephalogram) channel 4 module ID
5012	CO	CO module ID
5013	CO2	CO2 module ID
5014	N2O	N2O module ID
5015	O2	O2 module ID
5016	AA	AA module ID
5017	RM	RM(Respiratory mechanics) module ID
5021	TEMPA	TEMP(Temperature) A module ID
5022	TEMPB	TEMP(Temperature) B module ID
5023	TEMPC	TEMP(Temperature) C module ID
5024	TEMPD	TEMP(Temperature) D module ID
5025	TEMPE	TEMP(Temperature) E module ID
5026	TEMPF	TEMP(Temperature) F module ID
5027	TEMPG	TEMP(Temperature) G module ID
5031	IBP_P1	IBP(Invasive Blood Pressure) P1 module ID
5032	IBP_P2	IBP(Invasive Blood Pressure) P2 module ID
5033	IBP_P3	IBP(Invasive Blood Pressure) P3 module ID
5034	IBP_P4	IBP(Invasive Blood Pressure) P4 module ID
5035	IBP_P5	IBP(Invasive Blood Pressure) P5 module ID
5036	IBP_P6	IBP(Invasive Blood Pressure) P6 module ID
5037	IBP_P7	IBP(Invasive Blood Pressure) P7 module ID
5038	IBP_P8	IBP(Invasive Blood Pressure) P8 module ID

5039	IBP_ART	IBP(Invasive Blood Pressure) ART module ID
5040	IBP_CVP	IBP(Invasive Blood Pressure) CVP module ID
5041	IBP_PA	IBP(Invasive Blood Pressure) PA module ID
5042	IBP_RAP	IBP(Invasive Blood Pressure) RAP module ID
5043	IBP_LAP	IBP(Invasive Blood Pressure) LAP module ID
5044	IBP_ICP	IBP(Invasive Blood Pressure) ICP module ID
5045	IBP_UAP	IBP(Invasive Blood Pressure) UAP module ID
5046	IBP_BAP	IBP(Invasive Blood Pressure) BAP module ID
5047	IBP_FAP	IBP(Invasive Blood Pressure) FAP module ID
5048	IBP_UVP	IBP(Invasive Blood Pressure) UVP module ID
5049	IBP_LV	IBP(Invasive Blood Pressure) LV module ID
5050	IBP_AO	IBP(Invasive Blood Pressure) AO module ID
~~5051~~	~~TOF~~	
5052	CPP	CPP(Cerebral perfusion pressure) module ID
5053	RSO2_1	RSO2(cerebral and regional oxygen saturation) channel 1 module ID
5054	RSO2_2	RSO2(cerebral and regional oxygen saturation) channel 2 module ID
5055	RSO2_3	RSO2(cerebral and regional oxygen saturation) channel 3 module ID
5056	RSO2_4	RSO2(cerebral and regional oxygen saturation) channel 4 module ID
5057	NMT	NMT(Neuromuscular conduction) module ID
5058	EWS	EWS(Early Warning Score) module ID
5059	GCS	GCS(Glasgow Coma Scale) module ID
5060	DM	DM(infusion monitoring) module ID

Appendix A.4 Patient information ID

ID	Content	Describe	Value
4201	Height	Patient Height	
4202	Weight	Patient Weight	
4203	Blood Type	Patient Blood	0:Unknown 1:A 2:B 3:AB 4:O 5:A_ 6:B_ 7:AB_ 8:O_
4204	Pacing	Patient Paced	0:Off 1:On

			2:Auto 3:Unknown

Appendix A.5 Controls and settings

ID	Content	Describe
1001	Module online	Online module information（Module Load） Example: MSH|^~\&|||||20230518094209||ORU^R01^ORU_R01|1001|P|2.6||||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3||||||F OBX||NM|4204^Pace^BHC||2||||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A OBR||||Monitor|||20230518094209 OBX||CE|5^^BHC||5001^ECG^BHC||||||F MSH_10:1001 Indicates that the message is the information of all modules on the line; OBX||CE|5^^BHC||5001^ECG^BHC||||||F indicates that the ECG module is online. For details about the field, see *4.2.7.1 Online Module*.
1002	Module offline	Offline module information（Module Unload） Example: MSH|^~\&|||||20230518135240||ORU^R01^ORU_R01|1002|P|2.6||||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3||||||F OBX||NM|4204^Pace^BHC||2||||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF||||||||||||||A OBR||||Monitor|||20230518094209 OBX||CE|6^^BHC||5001^ECG^BHC||||||F MSH_10:1002 Indicates that the message is the information of all modules offline; In the OBX field, 6^^BHC indicates that the ECG module is offline. 5001^ECG^BHC indicates that the ECG module is offline.
1003	Supported (or unsupported parameters)	Parameter information supported (or not supported) by the module（Parameter Load And Unload） Example: MSH|^~\&|||||20230518135240||ORU^R01^ORU_R01|1003|P|2.6||||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3||||||F OBX||NM|4204^Pace^BHC||2||||||F

		PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518094209 OBX||CE|7^^BHC|5001|201^HR^BHC||||||F MSH_10:1003 Indicates that the message is all parameter information supported (or not supported) by the module; OBX||CE|7^^BHC|5001|201^HR^ The BHC||||||F field indicates that the ECG module supports the HR parameter. For details about the field, see *4.2.7.2 Supported Parameters*.
1004	Periodic parameters	Periodic parameters（Periodic Parameter） Example: MSH|^~\&||||||ORU^R01^ORU_R01|1004|P|2.6|||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20210914143443 OBX||NM|201^HR^BHC|5001|60|||||F OBX||NM|202^QTC^BHC|5001|359|||||F OBX||NM|203^DELTAQTC^BHC|5001|-1|||||F OBX||NM|204ST-I^BHC|5001|0.09|||||F OBX||NM|205ST-II^BHC|5001|0.21|||||F OBX||NM|206ST-III^BHC|5001|0.12|||||F OBX||NM|207ST-aVR^BHC|5001|-0.15|||||F ..... MSH_10:1004 Indicates that the message is all periodic parameter data information of all modules; OBX||NM|259^PR^BHC|5002|60|||||F indicates that the PR pulse rate of SpO2 is 60 bpm. For details about the field, see *4.2.7.3 Periodic Parameter Values.*
1005	Aperiodic parameter NIBP	Aperiodic parameters NIBP（Periodic Parameter_NIBP） Example: MSH|^~\&||||20230518135251||ORU^R01^ORU_R01|1005|P|2.6|||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518101324 OBX||NM|351^NIBP S^BHC|5004|125|11^mmHg^BHC|||||F||APERIODIC|20230518101323 OBX||NM|352^NIBP D^BHC|5004|84|11^mmHg^BHC|||||F||APERIODIC|20230518101323 OBX||NM|353^NIBP M^BHC|5004|96|11^mmHg^BHC|||||F||APERIODIC|20230518101323

		OBX||NM|354^NIBP SDP^BHC|5004|41|11^mmHg^BHC|||||F||APERIODIC|20230518101323 OBX||NM|355^NIBP PR^BHC|5004|60||||||F||APERIODIC|20230518101323 MSH_10:1005 Indicates that the message is non periodic NIBP parameter measurement data information OBX||NM|352^NIBP D^BHC|5004|84||||||F||APERIODIC|20230518101323 indicates that the systolic pressure of the aperiodic NIBP parameter is 84 mmHg. For details about the field, see *4.2.7.7 Aperiodic Parameter Values*.
1006	Aperiodic parameter CO	Aperiodic parameters CO（Periodic Parameter_CO） Example: MSH|^~\&|||||20230518135250||ORU^R01^ORU_R01|1006|P|2.6||||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3||||||F OBX||NM|4204^Pace^BHC||2||||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518101323 OBX||NM|751^C.O.^BHC|5012|5.2|||||F||APERIODIC|20230518101323 OBX||NM|752^C.I.^BHC|5012|2.9|||||F||APERIODIC|20230518101323 OBX||NM|753^TI^BHC|5012|-1|||||F||APERIODIC|20230518101323 OBX||NM|754^TB^BHC|5012|36.87|21^C^BHC||||F||APERIODIC|20230518101323 MSH_10:1006 Indicates that the message is aperiodic co parameter measurement data information； OBX||NM|751^C.O.^BHC|5012|5.2|||||F||APERIODIC|20230518101323 The field indicates that the cardiac output of the aperiodic parameter CO is 5.2 L/min. For details about the field, see *4.2.7.7 Aperiodic Parameter Values*.
1007	Physiological alarm	Physiological alarm information（Physiological Alarm） Example: MSH|^~\&|||||20230518135243||ORU^R01^ORU_R01|1007|P|2.6||||||UTF-8 PID|||||||M OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3||||||F OBX||NM|4204^Pace^BHC||2||||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518135243 OBX||CE|2||10151^SpO2 High^BHC|||||F|||20230608152545 OBX||CE|2||10202^NIBP-Sys Low^BHC|||||F|||20230608152441 OBX||CE|2||10206^NIBP-Dia Low^BHC|||||F|||20230608152441 MSH_10:1007 Indicates that the message is physiological alarm information of all triggered parameters；

		OBX||CE|2||10202^NIBP-Sys Low^BHC|||||F|||20230608152441 indicates that the physiological alarm occurs at 20230608152441 as the NIBP-Sys Low alarm. For details about the field, see *4.2.7.4 Physiological Alarm*.
1008	Technical alarm	Technical alarm information（Technical Alarm） Example: MSH|^~\&|||||20230518135252||ORU^R01^ORU_R01|1008|P|2.6|||||UTF-8 PID|||||||M OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518135252 OBX||CE|0||20307^IBP1 Zero Required^BHC|||||F|||20230518135242 OBX||CE|0||20327^IBP2 Zero Required^BHC|||||F|||20230518135242 MSH_10:1008 Indicates that the message is the technical alarm information of all modules triggered at this time； OBX||CE|0||20307^IBP1 Zero Required^BHC|||||F|||20230518135242 indicates that the technical alarm is IBP1 Zero Required at 20230518135242. For details about the fields, see *4.2.7.5 Technical Alarm*.
1009	Set alarm limit	Setting of alarm upper and lower limits（ Setup Limit） Example: MSH|^~\&|||||20230518160843||ORU^R01^ORU_R01|1009|P|2.6|||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518101318 OBX||NM|1^^BHC|201|120|||||F OBX||NM|2^^BHC|201|50|||||F OBX||NM|1^^BHC|251|100|||||F OBX||NM|2^^BHC|251|90|||||F MSH_10:1009 Indicates that the message is the setting information of the upper and lower limit of the alarm； OBX||NM|1^^BHC|251|100|||||F field indicates that the alarm upper limit set by parameter SPO2 is 100. For details about the field, see *4.2.7.8 Parameter Alarm limit*.
1010	Set alarm level	Setting of alarm upper and lower limits（Alarm Setup Level） Example: MSH|^~\&|||||20230518135233||ORU^R01^ORU_R01|1010|P|2.6|||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F

		OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR|||Monitor|||20230518101318 OBX||NM|3^^BHC|201|2|||||F OBX||NM|3^^BHC|251|2|||||F OBX||NM|3^^BHC|351|2|||||F MSH_10:1010 Indicates that the message is the setting information of alarm level; OBX||NM|3^^BHC|351|2|||||F indicates that the diastolic pressure alarm level set by NIBP is intermediate alarm level. For details about the field, see *4.2.7.9 Parameter Alarm Level*.
1011	Set alarm switch	Setting of alarm upper and lower limits（Alarm Setup Switch） Example: MSH|^~\&||||20230518135241||ORU^R01^ORU_R01|1011|P|2.6|||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR|||Monitor|||20230518101318 OBX||NM|4^^BHC|201|1|||||F OBX||NM|4^^BHC|251|1|||||F OBX||NM|4^^BHC|351|1|||||F MSH_10:1011 Indicates that the message is the setting information of the alarm switch; OBX||NM|4^^BHC|351|1|||||F indicates that the alarm switch for diastolic pressure setting of NIBP is on. For details about the field, see *4.2.7.10 Parameter alarm switch*.
1012	Broadcast machine information	Broadcast machine information Example: MSH|^~\&||||20230523111421||ADT^A01^ADT_A01|1012|P|2.6|||||UTF-8 EVN||20230512094930 PID|||78||Mis^Lucy^Merry||20020202|F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBX||ST|4002^Monitor Name^BHC||Monitor1|||||F MSH_10:1012 indicates that the message is a machine message broadcast by the monitor; OBX||ST|4002^Monitor Name^BHC||Monitor1|||||F The field indicates the monitor name is Monitor1 and is described in section *6.2.6 OBX Segment*.
1014	TCP Heartbeat	Heartbeat Packet Message（Heartbeat Packet） Example:

		MSH|^~\&|||||20230614102744||ORU^R01^ORU_R01|1014|P|2.6||||||UTF-8 MSH_10:1014 indicates that the message is a Tcp heartbeat packet.
1015	Waveform data	Waveform Data Example: MSH|^~\&|||||20230518135241||ORU^R01^ORU_R01|1015|P|2.6||||||UTF-8 MSH_10:1015 indicates that the message is a waveform data message; The waveform data contains a lot of information. For details, see *4.3 Waveform Data Sending*.
1016	ADT Patient Query Message	ADT Patient Query Message(QRY_A19) Example: MSH|^~\\&||||ADTServer|20220906095605||QRY^A19^QRY_A19|1016|P|2.6||||||UTF-8 QRD|20220906095605|R|I|2|||1RD|M2021|^DEM|^BLTCMS MSH_10:1016 indicates that the message is an ADT patient query message; QRD_8: M2021 in the field is the patient's medical record number (patient's unique identifier). See section *7.2.4 QRD Segment* for detailed description of the field.
1017	ADT Patient Query Result Message	ADT Patient Query Result Message(ADR_A19) Example: MSH|^~\\&||||20220906111243||ADR^A19^ADR_A19|1017|P|2.6||||||UTF-8 MSA|AA|1016|Query Found QRD|20220906095605|R|I|2|||1RD|M2021|^DEM|^BLTCMS PID|||M2021||Mis^Luck^Bob||19951108|F PV1||I|CCU^R1^B4&36&0|||DoctorInfo||||||||||U OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F OBX||NM|4203^Blood^BHC||3||||||F OBX||NM|4204^Pace^BHC||2||||||F MSH_10:1017 indicates that the message is a query result message for ADT patients; The message returns patient information (patient record number, name, date of birth, etc.)
1018	ACK Acknowledge Message	ACK Acknowledge Message Example: MSH|^~\\&||||20220906115151||ACK|1018|P|2.6||||||UTF-8 MSA|AA|1016 MSH_10:1018 indicates that the message is a message confirmation message.
1025	Aperiodic parameter NMT	Aperiodic parameter NMT（Periodic Parameter_NMT） Example: MSH|^~\&|||||20230518135241||ORU^R01^ORU_R01|1025|P|2.6||||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC|||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC|||||F

		OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518115149 OBX||ST|2201^Stimulate Current^BHC|5057|25^25^25^25|||||F||APERIODIC|20230518115149 OBX||ST|2202^Measure Current^BHC|5057|24.80^24.81^24.80^24.79|||||F||APERIODIC|20230518115149 OBX||ST|2203^Bargraph Strength^BHC|5057|70^10^10^0|||||F||APERIODIC|20230518115149 OBX||NM|2204^Measure Result^BHC|5057|3|||||F||APERIODIC|20230518115149 OBX||NM|2205^Result Type^BHC|5057|0|||||F||APERIODIC|20230518115149 OBX||NM|2206^Stimulate Cycle^BHC|5057|-1|||||F||APERIODIC|20230518115149 OBX||NM|2207^Stimulate Mode^BHC|5057|3|||||F||APERIODIC|20230518115149 OBX||NM|2208^Auto Mode^BHC|5057|-1|||||F||APERIODIC|20230518115149 OBX||NM|2209^Bargraph Number^BHC|5057|4|||||F||APERIODIC|20230518115149 MSH_10:1025 indicates that the message is aperiodic NMT parameter measurement data; OBX||ST|2202^Measure Current^BHC|5057|24.80^24.81^24.80^24.79|||||F||APERIODIC|20230518115149 indicates that the four measured currents of the NMT parameter are 24.80, 24.81, 24.80 and 24.79 mA respectively.
1026	Aperiodic parameter PAWP	Aperiodic parameter PAWP (Periodic Parameter_PAWP) Example: MSH|^~\&||||20230518135241||ORU^R01^ORU_R01|1026|P|2.6|||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518114729 OBX||NM|1706^Pawp^BHC|5041|81|11^mmHg^BHC||||F||APERIODIC|20230518114729 MSH_10:1026 indicates that the message is aperiodic PAWP parameter measurement data information; OBX||NM|1706^Pawp^BHC|5041|81|||||F||APERIODIC|20230518114729 indicates that the Pawp parameter value in the IBP_PA module is 81 mmHg. For details about the field, see *4.2.7.7 Aperiodic Parameter Value*.
1027	Aperiodic parameter EWS	Aperiodic parameter EWS (Periodic Parameter_EWS) Example: MSH|^~\&||||20230518135241||ORU^R01^ORU_R01|1027|P|2.6|||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A

		OBR||||Monitor|||20230518135302 OBX||NM|2051^Total Score^BHC|5058|5|||||F||APERIODIC|20230518135302 OBX||NM|2052^Score Type^BHC|5058|0|||||F||APERIODIC|20230518135302 MSH_10:1027 indicates that the message is aperiodic EWS parameter measurement data; OBX||NM|2051^Total Score^BHC|5058|5|||||F||APERIODIC|20230518135302 indicates that the Total Score of the EWS module is 5. For details about the fields, see *4.2.7.7 Aperiodic Parameter Values*.
1028	Aperiodic parameter GCS	Aperiodic parameter GCS (Periodic Parameter_GCS) Example: MSH|^~\&||||20230518135241||ORU^R01^ORU_R01|1028|P|2.6|||||UTF-8 PID|||78||Mis^Lucy^Merry||20020202|F OBX||NM|4201^Height^BHC||170.0|31^cm^BHC||||F OBX||NM|4202^Weight^BHC||65.0|41^kg^BHC||||F OBX||NM|4203^Blood^BHC||3|||||F OBX||NM|4204^Pace^BHC||2|||||F PV1||I|^^&8&0^^^^^^1CBA8CB36CEF|||||||||||||A OBR||||Monitor|||20230518135251 OBX||NM|2101^Total Score^BHC|5059|9|||||F||APERIODIC|20230518135251 OBX||NM|2102^Eye Opening^BHC|5059|2|||||F||APERIODIC|20230518135251 OBX||NM|2103^Verbal Response^BHC|5059|3|||||F||APERIODIC|20230518135251 OBX||NM|2104^Motor Response^BHC|5059|4|||||F||APERIODIC|20230518135251 MSH_10:1028 indicates that the message is aperiodic GCS parameter measurement data information; OBX||NM|2102^Eye Opening^BHC|5059|2|||||F||APERIODIC|20230518135251 indicates that the single score of the Eye Opening parameter of the GCS module is 2. For details about the fields, see *4.2.7.7 Aperiodic Parameter Values*.
1029	Realtime Query Message	Realtime Query Message（QRY_R02） Example: MSH|^~\&||||20220424095857||ORY^R02^ORY_R02|1029|P|2.6|||||UTF-8 QRD|20220424095857|R|I|2||||RES QRF|MON||||0&0^1^0^0 MSH_10:1029 indicates that the message is real-time query information; QRD|20220424095857|R|I|2||||RES, see section 5.2.4 QRD for details; QRF|MON||||0&0^1^0^0 In the Segment, 1 indicates that the type of information queried is physiological parameter data. For details about the fields, see section *5.2.5 QRF Segment*.
1030	Realtime Query Result Message	Realtime Query Result Message（ORF_R04） Examlpe: MSH|^~\\&||||20230616115450||ORF^R04^ORF_R04|1030|P|2.6|||||UTF-8 MSA|AA|1029 QRD|20230616115452|R|I|2||||RES

		QRF|MON||||0&0^2^0^0 PID|||78||Mis^Lucy^Merry||20020202|F OBR||||Monitor|||20230616115450 OBX||CE|2||10151^SpO2 High^BHC|||||F|||20230616115207 OBX||CE|2||10202^NIBP-Sys Low^BHC|||||F|||20230616115218 OBX||CE|2||10206^NIBP-Dia Low^BHC|||||F|||20230616115218 OBX||CE|2||10203^NIBP-Map High^BHC|||||F|||20230616115218 OBX||CE|2||10208^NIBP-Sdp Low^BHC|||||F|||20230616115218 MSH_10:1030 indicates that the message is a real-time query result; The example is a real-time result message requesting the physiological alarm of the monitor. QRF_5:0&0^2^0^0 The binary code of 2 in the field is 0010, indicating that the requested data type is physiological alarm. For details about the message, see *5.2.5 QRF Segment*.

Appendix A.6 Physiological Alarm ID

ID	Content
10001	HR High
10002	HR Low
10003	HR Too High
10004	HR Too Low
10005	ECG Weak signal
10011	QTc High
10012	Delta QTc High
10021	ST_I High
10022	ST_I Low
10023	ST_IIHigh
10024	ST_II Low
10025	ST_III High
10026	ST_III Low
10027	ST_AVR High
10028	ST_AVR Low
10029	ST_AVL High
10030	ST_AVL Low
10031	ST_AVF High
10032	ST_AVF Low
10033	ST_V1 High
10034	ST_V1 Low
10035	ST_V2 High
10036	ST_V2 Low
10037	ST_V3 High

10038	ST_V3 Low
10039	ST_V4 High
10040	ST_V4 Low
10041	ST_V5 High
10042	ST_V5 Low
10043	ST_V6 High
10044	ST_V6 Low
10045	ST_V High
10046	ST_V Low
10047	ST_Va High
10048	ST_Va Low
10049	ST_Vb High
10050	ST_Vb Low
10101	ECG Asystole
10102	ECG Vent Fib/Tach
10103	ECG Vent Brady
10104	ECG Extreme Tachy
10105	ECG Extreme Brady
10106	ECG V-Tach
10107	ECG Run PVCs
10108	ECG Couplet
10109	ECG Bigeminy
10110	ECG Trigeminy
10111	ECG R On T
10112	ECG PVC
10113	ECG Tachy
10114	ECG Brady
10115	ECG Missed Beat
10116	ECG A-Fib
10117	ECG A-Fib End
10118	ECG PNP
10119	ECG PNC
10120	ECG Noise
10121	ECG Frequent PVCs
10122	ECG Nonsustained V-Tach
10123	ECG Vent Rhythm
10124	ECG Pause
10125	ECG Irregular Rhythm
10126	ECG Pauses/min High
10127	ECG Irregular Rhythm End
10128	ECG Learning
10151	SPO2 High
10152	SPO2 Low

10153	SPO2 Desat
10154	Delta SPO2 High
10155	SPO2 Search Pulse Timeout
10156	SPO2 RR High
10157	SPO2 RR Low
10158	SPO2 Apnea
10159	SPO2 PI Low
10171	SPO2L High
10172	SPO2L Low
10173	SPO2L Desat
10174	Delta SPO2 High
10175	SPO2L Search Pulse Timeout
10176	SPO2L RR High
10177	SPO2L RR Low
10178	SPO2L Apnea
10179	SPO2L PI Low
10201	NIBP SYS High
10202	NIBP SYS Low
10203	NIBP MAP High
10204	NIBP MAP Low
10205	NIBP DIA High
10206	NIBP DIA Low
10207	NIBP SDP High
10208	NIBP SDP Low
10251	RESP RR High
10252	RESP RR Low
10253	RESP Apnea
10254	RESP Artifact
10301	IBP1 SYS High
10302	IBP1 SYS Low
10303	IBP1 MEAN High
10304	IBP1 MEAN Low
10305	IBP1 DIA High
10306	IBP1 DIA Low
10311	IBP2 SYS High
10312	IBP2 SYS Low
10313	IBP2 MEAN High
10314	IBP2 MEAN Low
10315	IBP2 DIA High
10316	IBP2 DIA Low
10321	IBP3 SYS High
10322	IBP3 SYS Low
10323	IBP3 MEAN High

10324	IBP3 MEAN Low
10325	IBP3 DIA High
10326	IBP3 DIA Low
10331	IBP4 SYS High
10332	IBP4 SYS Low
10333	IBP4 MEAN High
10334	IBP4 MEAN Low
10335	IBP4 DIA High
10336	IBP4 DIA Low
10341	IBP5 SYS High
10342	IBP5 SYS Low
10343	IBP5 MEAN High
10344	IBP5 MEAN Low
10345	IBP5 DIA High
10346	IBP5 DIA Low
10351	IBP6 SYS High
10352	IBP6 SYS Low
10353	IBP6 MEAN High
10354	IBP6 MEAN Low
10355	IBP6 DIA High
10356	IBP6 DIA Low
10361	IBP7 SYS High
10362	IBP7 SYS Low
10363	IBP7 MEAN High
10364	IBP7 MEAN Low
10365	IBP7 DIA High
10366	IBP7 DIA Low
10371	IBP8 SYS High
10372	IBP8 SYS Low
10373	IBP8 MEAN High
10374	IBP8 MEAN Low
10375	IBP8 DIA High
10376	IBP8 DIA Low
10381	IBP_ART SYS High
10382	IBP_ART SYS Low
10383	IBP_ART MEAN High
10384	IBP_ART MEAN Low
10385	IBP_ART DIA High
10386	IBP_ART DIA Low
10391	IBP_CVP SYS High
10392	IBP_CVP SYS Low

10393	IBP_CVP MEAN High
10394	IBP_CVP MEAN Low
10395	IBP_CVP DIA High
10396	IBP_CVP DIA Low
10401	PA SYS High
10402	PA SYS Low
10403	PA MEAN High
10404	PA MEAN Low
10405	PA DIA High
10406	PA DIA Low
10411	RAP SYS High
10412	RAP SYS Low
10413	RAP MEAN High
10414	RAP MEAN Low
10415	RAP DIA High
10416	RAP DIA Low
10421	LAP SYS High
10422	LAP SYS Low
10423	LAP MEAN High
10424	LAP MEAN Low
10425	LAP DIA High
10426	LAP DIA Low
10431	ICP SYS High
10432	ICP SYS Low
10433	ICP MEAN High
10434	ICP MEAN Low
10435	ICP DIA High
10436	ICP DIA Low
10441	UAP SYS High
10442	UAP SYS Low
10443	UAP MEAN High
10444	UAP MEAN Low
10445	UAP DIA High
10446	UAP DIA Low
10451	BAP SYS High
10452	BAP SYS Low
10453	BAP MEAN High
10454	BAP MEAN Low
10455	BAP DIA High
10456	BAP DIA Low

10461	FAP SYS High
10462	FAP SYS Low
10463	FAP MEAN High
10464	FAP MEAN Low
10465	FAP DIA High
10466	FAP DIA Low
10471	UVP SYS High
10472	UVP SYS Low
10473	UVP MEAN High
10474	UVP MEAN Low
10475	UVP DIA High
10476	UVP DIA Low
10481	LV SYS High
10482	LV SYS Low
10483	LV MEAN High
10484	LV MEAN Low
10485	LV DIA High
10486	LV DIA Low
10491	AO SYS High
10492	AO SYS Low
10493	AO MEAN High
10494	AO MEAN Low
10495	AO DIA High
10496	AO DIA Low
10601	ETCO2 High
10602	ETCO2 Low
10603	FICO2 High
10604	FICO2 Low
10605	AWRR High
10606	AWRR Low
10607	CO2 Apnea
10611	ETO2 High
10612	ETO2 Low
10613	FIO2 High
10614	FIO2 Low
10621	ETN2O High
10622	ETN2O Low
10623	FIN2O High
10624	FIN2O Low
10631	Agent_No_ETAA High

10632	Agent_No_ETAA Low
10633	Agent_No_FIAA High
10634	Agent_No_FIAA Low
10641	Agent_Halothane_ETAA High
10642	Agent_Halothane_ETAA Low
10643	Agent_Halothane_FIAA High
10644	Agent_Halothane_FIAA Low
10651	Agent_Enflurane_ETAA High
10652	Agent_Enflurane_ETAA Low
10653	Agent_Enflurane_FIAA High
10654	Agent_Enflurane_FIAA Low
10661	Agent_Isoflurane_ETAA High
10662	Agent_Isoflurane_ETAA Low
10663	Agent_Isoflurane_FIAA High
10664	Agent_Isoflurane_FIAA Low
10671	Agent_Sevoflurane_ETAA High
10672	Agent_Sevoflurane_ETAA Low
10673	Agent_Sevoflurane_FIAA High
10674	Agent_Sevoflurane_FIAA Low
10681	Agent_Desflurane_ETAA High
10682	Agent_Desflurane_ETAA Low
10683	Agent_Desflurane_FIAA High
10684	Agent_Desflurane_FIAA Low
10691	PR High
10692	PR Low
10701	CO-TB High
10702	CO-TB Low
10751	RM-PEEP High
10752	RM-PEEP Low
10753	RM-PIP High
10754	RM-PIP Low
10755	RM-MVE High
10756	RM-MVE Low
10757	RM-RR High
10758	RM-RR Low
10759	RM Apnea
10781	CPP High
10782	CPP Low
10901	ICG-C.I High
10902	ICG-C.I Low

10903	ICG-TFC High
10904	ICG-TFC Low
10951	BIS High
10952	BIS Low
11001	RSO2-CH1 High
11002	RSO2-CH1 Low
11003	RSO2-CH1-DeltaSpo2 High
11004	RSO2-CH2 High
11005	RSO2-CH2 Low
11006	RSO2-CH2-DeltaSpo2 High
10801	T1 High
10802	T1 Low
10803	T2 High
10804	T2 Low
10805	T3 High
10806	T3 Low
10807	T4 High
10808	T4 Low
10809	T5 High
10810	T5 Low
10811	T6 High
10812	T6 Low
10813	T7 High
10814	T7 Low
10815	T8 High
10816	T8 Low
10817	T9 High
10818	T9 Low
10819	T10 High
10820	T10 Low
10821	Skin Temp High
10822	Skin Temp Low
10823	Core Temp High
10824	Core Temp Low
10825	Axil Temp High
10826	Axil Temp Low
10827	Naso Temp High
10828	Naso Temp Low
10829	Eso Temp High
10830	Eso Temp Low

10831	Rect Temp High
10832	Rect Temp Low
10833	Amb Temp High
10834	Amb Temp Low
10835	Airw Temp High
10836	Airw Temp Low
10837	Vesic Temp High
10838	Vesic Temp Low
10839	Blood Temp High
10840	Blood Temp Low
10841	Myo Temp High
10842	Myo Temp Low
10843	Tymp Temp High
10844	Tymp Temp Low
10845	Cereb Temp High
10846	Cereb Temp Low
10847	TA High
10848	TA Low
10849	TB High
10850	TB Low
10851	TC High
10852	TC Low
10853	TD High
10854	TD Low
10855	TE High
10856	TE Low

Appendix A.7 Technical alarm ID

ID	Content
20001	ECG Communication Error
20002	ECG Self Checking Error
20003	ECG V1 Sensor Off
20004	ECG V2 Sensor Off
20005	ECG V3 Sensor Off
20006	ECG V4 Sensor Off
20007	ECG V5 Sensor Off
20008	ECG V6 Sensor Off
20009	ECG Ra Sensor Off

20010	ECG RL Sensor Off
20011	ECG La Sensor Off
20012	ECG LL Sensor Off
20013	ECG V Sensor Off
20014	ECG Va Sensor Off
20015	ECG Vb Sensor Off
20016	ECG Leads Sensor Off
20017	ECG V1 Polarized
20018	ECG V2 Polarized
20019	ECG V3 Polarized
20020	ECG V4 Polarized
20021	ECG V5 Polarized
20022	ECG V6 Polarized
20023	ECG I Polarized
20024	ECG II Polarized
20025	ECG III Polarized
20026	ECG AVR Polarized
20027	ECG AVL Polarized
20028	ECG AVF Polarized
20029	ECG Va Polarized
20030	ECG Vb Polarized
20031	ECG can't analyse ST
20032	ECG can't analyse QT
20033	ECG Suspected Pacing Signal
20034	ECG Cable Will Expire Soon
20051	SPO2 Communication Error
20052	SPO2 Self Checking Error
20053	SPO2 Sensor Off
20054	SPO2 Searching Pulse...
20055	SPO2 Low Confidence
20056	SPO2 No Sensor Connected
20057	SPO2 Value Update Timeout
20058	SPO2 PR Value Update Timeout
20059	SPO2 No Cable
20060	SPO2 Replace Cable
20061	SPO2 Cable validation failed
20062	SPO2 Unrecognized Cable
20063	SPO2 TBD
20064	SPO2 No Sensor
20065	SPO2 Invalid Sensor
20066	SPO2 Replace Sensor
20067	SPO2 Motion Interference
20068	SPO2 Sensor Incompatible

20069	SPO2 Check Sensor
20070	SPO2 DEMO TOOL
20071	SPO2 Low Perfusion
20072	SPO2 Sensor Initializing
20073	SPO2 Replace Adhesive
20074	SPO2 No Adhesive
20075	SPO2 Incompatible Adhesive
20076	SPO2 Unrecognized Adhesive
20077	SPO2 PI Low Confidence
20078	SPO2 PVI Low Confidence
20079	SPO2 SpCO Low Confidence
20080	SPO2 SpOC Low Confidence
20081	SPO2 SpMet Low Confidence
20082	SPO2 SpHb Low Confidence
20101	SPO2LCommunication Error
20102	SPO2LSelf Checking Error
20103	SPO2L Sensor Off
20104	SPO2L Pulse Search
20105	SPO2L Low Confidence
20106	SPO2L No Sensor Connected
20107	SPO2L Value Update Timeout
20108	SPO2L PR Value Update Timeout
20109	SPO2L No Cable
20110	SPO2L Replace Cable
20111	SPO2L Cable validation failed
20112	SPO2L Unrecognized Cable
20113	SPO2L TBD
20114	SPO2L No Sensor
20115	SPO2L Invalid Sensor
20116	SPO2L Replace Sensor
20117	SPO2L Motion Interference
20118	SPO2L Sensor Incompatible
20119	SPO2L Check Sensor
20120	SPO2L DEMO TOOL
20121	SPO2L Low Perfusion
20122	SPO2L Sensor Initializing
20123	SPO2L Replace Adhesive
20124	SPO2L No Adhesive
20125	SPO2L Incompatible Adhesive
20126	SPO2 Unrecognized Adhesive
20127	SPO2L PI Low Confidence
20128	SPO2L PVI Low Confidence
20129	SPO2L SpCO Low Confidence

20130	SPO2L SpOC Low Confidence
20131	SPO2L SpMet Low Confidence
20132	SPO2L SpHb Low Confidence
20151	NIBP Communication Error
20152	NIBP Hardware Self Checking Error
20153	NIBP Self Checking Error
20154	NIBP Loose Cuff
20155	NIBP Air Leakage
20156	NIBP Air Pressure Error
20157	NIBP Weak signal
20158	NIBP Out of Range
20159	NIBP Excessive motion
20160	NIBP Overpressure Detected
20161	NIBP Signal Saturated
20162	NIBP Air System Leak
20163	NIBP System Failure
20164	NIBP Time out
20165	NIBP Cuff Type Error
20166	NIBP Zero Failed
20167	NIBP Calibration Failed
20168	NIBP Overpressure Zero Failed
20169	NIBP Overpressure Calibrate Failed
20170	NIBP Cycle Abort
20171	NIBP Exceeded Retry Count
20172	NIBP Pneumatic Blockage
20173	NIBP Measurement Terminated
20174	NIBP Air Leak or Loose Cuff
20175	NIBP Safety Timeout
20176	NIBP Calibration Successful
20177	NIBP Calibration Failure
20201	RESP Communication Error
20202	RESP Self-test Failed
20203	RESP Sensor Off
20301	IBP1 Communication Error
20302	IBP1 Self Checking Error
20303	IBP1 Calibration Failed
20304	IBP1 Zero Failed
20305	IBP1 Sensor Off
20306	IBP1 Catheter Off
20307	IBP1 Zero Required
20308	IBP1 Zero In Progress...
20309	IBP1 Zero Succeed
20310	IBP1 Pressure Calibrating

20311	IBP1 Pressure Calibration Succeed
20321	IBP2 Communication Error
20322	IBP2 Self Checking Error
20323	IBP2 Calibration Failed
20324	IBP2 Zero Failed
20325	IBP2 Sensor Off
20326	IBP2 Catheter Off
20327	IBP2 Zero Required
20328	IBP2 Zero In Progress...
20329	IBP2 Zero Succeed
20330	IBP2 Pressure Calibrating
20331	IBP2 Pressure Calibration Succeed
20341	IBP3 Communication Error
20342	IBP3 Self Checking Error
20343	IBP3 Calibration Failed
20344	IBP3 Zero Failed
20345	IBP3Sensor Off
20346	IBP3 Catheter Off
20347	IBP3 Zero Required
20348	IBP3 Zero In Progress...
20349	IBP3 Zero Succeed
20350	IBP3 Pressure Calibrating
20351	IBP3 Pressure Calibration Succeed
20361	IBP4 Communication Error
20362	IBP4 Self Checking Error
20363	IBP4 Calibration Failed
20364	IBP4 Zero Failed
20365	IBP4Sensor Off
20366	IBP4 Catheter Off
20367	IBP4 Zero Required
20368	IBP4 Zero In Progress...
20369	IBP4 Zero Succeed
20370	IBP4 Pressure Calibrating
20371	IBP4 Pressure Calibration Succeed
20381	IBP5 Communication Error
20382	IBP5 Self Checking Error
20383	IBP5 Calibration Failed
20384	IBP5 Zero Failed
20385	IBP5Sensor Off
20386	IBP5 Catheter Off
20387	IBP5 Zero Required
20388	IBP5 Zero In Progress...
20389	IBP5 Zero Succeed

20390	IBP5 Pressure Calibrating
20391	IBP5 Pressure Calibration Succeed
20401	IBP6 Communication Error
20402	IBP6 Self Checking Error
20403	IBP6 Calibration Failed
20404	IBP6 Zero Failed
20405	IBP6 Sensor Off
20406	IBP6 Catheter Off
20407	IBP6 Zero Required
20408	IBP6 Zero In Progress...
20409	IBP6 Zero Succeed
20410	IBP6 Pressure Calibrating
20411	IBP6 Pressure Calibration Succeed
20421	IBP7 Communication Error
20422	IBP7 Self Checking Error
20423	IBP7 Calibration Failed
20424	IBP7 Zero Failed
20425	IBP7 Sensor Off
20426	IBP7 Catheter Off
20427	IBP7 Zero Required
20428	IBP7 Zero In Progress...
20429	IBP7 Zero Succeed
20430	IBP7 Pressure Calibrating
20431	IBP7 Pressure Calibration Succeed
20441	IBP8 Communication Error
20442	IBP8 Self Checking Error
20443	IBP8 Calibration Failed
20444	IBP8 Zero Failed
20445	IBP8 Sensor Off
20446	IBP8 Catheter Off
20447	IBP8 Zero Required
20448	IBP8 Zero In Progress...
20449	IBP8 Zero Succeed
20450	IBP8 Pressure Calibrating
20451	IBP8 Pressure Calibration Succeed
20461	ART Communication Error
20462	ART Self Checking Error
20463	ART Calibration Failed
20464	ART Zero Failed
20465	ART Sensor Off
20466	ART Catheter Off
20467	ART Zero Required
20468	ART Zero In Progress...

20469	ART Zero Succeed
20470	ART Pressure Calibrating
20471	ART Pressure Calibration Succeed
20481	CVP Communication Error
20482	CVP Self Checking Error
20483	CVP Calibration Failed
20484	CVP Zero Failed
20485	CVP Sensor Off
20486	CVP Catheter Off
20487	CVP Zero Required
20488	CVP Zero In Progress...
20489	CVP Zero Succeed
20490	CVP Pressure Calibrating
20491	CVP Pressure Calibration Succeed
20501	PA Communication Error
20502	PA Self Checking Error
20503	PA Calibration Failed
20504	PA Zero Failed
20505	PA Sensor Off
20506	PA Catheter Off
20507	PA Zero Required
20508	PA Zero In Progress...
20509	PA Zero Succeed
20510	PA Pressure Calibrating
20511	PA Pressure Calibration Succeed
20521	RAP Communication Error
20522	RAP Self Checking Error
20523	RAP Calibration Failed
20524	RAP Zero Failed
20525	RAP Sensor Off
20526	RAP Catheter Off
20527	RAP Zero Required
20528	RAP Zero In Progress...
20529	RAP Zero Succeed
20530	RAP Pressure Calibrating
20531	RAP Pressure Calibration Succeed
20541	LAP Communication Error
20542	LAP Self Checking Error
20543	LAP Calibration Failed
20544	LAP Zero Failed
20545	LAP Sensor Off
20546	LAP Catheter Off
20547	LAP Zero Required

20548	LAP Zero In Progress...
20549	LAP Zero Succeed
20540	LAP Pressure Calibrating
20551	LAP Pressure Calibration Succeed
20561	ICP Communication Error
20562	ICP Self Checking Error
20563	ICP Calibration Failed
20564	ICP Zero Failed
20565	ICP Sensor Off
20566	ICP Catheter Off
20567	ICP Zero Required
20568	ICP Zero In Progress...
20569	ICP Zero Succeed
20570	ICP Pressure Calibrating
20571	ICP Pressure Calibration Succeed
20581	UAP Communication Error
20582	UAP Self Checking Error
20583	UAP Calibration Failed
20584	UAP Zero Failed
20585	UAP Sensor Off
20586	UAP Catheter Off
20587	UAP Zero Required
20588	UAP Zero In Progress...
20589	UAP Zero Succeed
20590	UAP Pressure Calibrating
20591	UAP Pressure Calibration Succeed
20601	BAP Communication Error
20602	BAP Self Checking Error
20603	BAP Calibration Failed
20604	BAP Zero Failed
20605	BAP Sensor Off
20606	BAP Catheter Off
20607	BAP Zero Required
20608	BAP Zero In Progress...
20609	BAP Zero Succeed
20610	BAP Pressure Calibrating
20611	BAP Pressure Calibration Succeed
20621	FAP Communication Error
20622	FAP Self Checking Error
20623	FAP Calibration Failed
20624	FAP Zero Failed
20625	FAP Sensor Off
20626	FAP Catheter Off

20627	FAP Zero Required
20628	FAP Zero In Progress...
20629	FAP Zero Succeed
20630	FAP Pressure Calibrating
20631	FAP Pressure Calibration Succeed
20641	UVP Communication Error
20642	UVP Self Checking Error
20643	UVP Calibration Failed
20644	UVP Zero Failed
20645	UVP Sensor Off
20646	UVP Catheter Off
20647	UVP Zero Required
20648	UVP Zero In Progress...
20649	UVP Zero Succeed
20650	UVP Pressure Calibrating
20651	UVP Pressure Calibration Succeed
20661	LV Communication Error
20662	LV Self Checking Error
20663	LV Calibration Failed
20664	LV Zero Failed
20665	LV Sensor Off
20666	LV Catheter Off
20667	LV Zero Required
20668	LV Zero In Progress...
20669	LV Zero Succeed
20670	LV Pressure Calibrating
20671	LV Pressure Calibration Succeed
20681	AO Communication Error
20682	AO Self Checking Error
20683	AO Calibration Failed
20684	AO Zero Failed
20685	AO Sensor Off
20686	AO Catheter Off
20687	AO Zero Required
20688	AO Zero In Progress...
20689	AO Zero Succeed
20690	AO Pressure Calibrating
20691	AO Pressure Calibration Succeed
20901	CO2 Communication Error
20902	CO2 Self Checking Error
20903	CO2 Sensor Off
20904	CO2 Check Adapter:E1
20905	CO2 Out of Range

20906	CO2 Sleep Mode
20907	CO2 Zero In Progress...
20908	CO2 Sensor Not Ready to Zero
20909	CO2 Zero Required
20910	CO2 Zero Required:E1
20911	CO2 Check Adapter:E2
20912	CO2 Sensor Hotting
20913	CO2 Sensor Over Temp
20914	CO2 temperature Is Not Stable
20915	CO2 Compensation Not Set
20916	CO2 Wait for Sensor
20917	CO2 Sensor Faulty:E1
20918	CO2 Sensor Faulty:E2
20919	CO2 EEPROM Checksum Error
20920	CO2 Pump Off
20921	CO2 Check Sample Line
20922	Pump Life expired
20923	Sample Line Disconnected
20924	CO2 Self-Test
20925	CO2 Outside Specified Accuracy Range
20926	CO2 Sensor Faulty:E5
20927	CO2 Motor Speed Error
20928	CO2 Factory calibration lost
20929	CO2 Replace adapter
20930	CO2 Sampling Line Clogged
20931	CO2 No adapter
20932	CO2 No sample line
20933	CO2 Ambient Pressure Outside Range
20934	CO2 Internal Temp Out of Range
20935	CO2 Span Calibration Command Failed
20936	CO2 Pressure Calibrating
20937	CO2 Calculation Overflow Detected
20938	CO2 temperature Is Not Stable
20939	CO2 System In Self_Zero Mode
20940	CO2 Sensor Temperature Low
20941	CO2 Sensor Temperature High
20942	CO2 Calibration Error Occurred
20943	CO2 ADC Voltage Error
20944	CO2 ADC Data Invalid
20945	CO2 Flowrate Out Of Range
20946	CO2 RAM Self-test Failed
20947	CO2 Main Code Self-test Failure
20948	CO2 Boot Code Self-test Failure

20949	CO2 Invalid command
20971	O2 Communication Error
20972	O2 Sensor Error
20973	O2 Interface error
20974	O2 Out of Range
20981	N2O Communication Error
20982	N2O Out of Range
20991	AA Communication Error
20992	AA Agent ID Are Unreliable
20993	AA At Least One Anesthetic Gas Out of Range
20994	AA Mixed Anesthetic Gas
21001	C.O. Communication Error
21003	C.O. TI Self Checking Error
21004	C.O. TB Self Checking Error
21005	C.O. Catheter Const Error
21006	C.O. Out of Range
21007	C.O. TB Unstable
21008	C.O. Measure Time Out
21009	C.O. TI Sensor Off
21010	C.O. TB Sensor Off
21011	C.O. TI Over High Range
21013	C.O. TI TI Below Low Range
21012	C.O. TB Over High Range
21014	C.O. TB TI Below Low Range
21051	RM Communication Error
21052	RM Sensor Off
21053	RM Check Zero Failed
21054	RM Power Err
21401	ICG Communication Error
21402	ICG Self Checking Error
21403	ICG Sensor Off
21404	ICG Electrode Off
21405	ICG Sensor Initializing 中
21406	ICG Check Sensor
21407	ICG Invalid Patient Settings
21408	ICG Invalid Blood Pressure Settings
21409	ICG External Measurement Settings
21410	ICG Invalid Hardware
21451	BIS Communication Error
21452	BIS Sensor Off
21453	BIS Bad SQI
21454	BIS Poor SQI
21455	BIS Simulator Connected

21601	EEG Communication Error
21602	EEG CH1 Electrode Off
21603	EEG CH2 Electrode Off
21604	EEG CH3 Electrode Off
21605	EEG CH4 Electrode Off
21606	EEG CH1 Polarized
21607	EEG CH2 Polarized
21608	EEG CH3 Polarized
21609	EEG CH4 Polarized
21610	EEG CH1 Impedance High
21611	EEG CH2 Impedance High
21612	EEG CH3 Impedance High
21613	EEG CH4 Impedance High
21651	DM Communication Error
21652	DM Drip Rate Abnormal
21653	DM Finished
21701	NMT Communication Error
21702	NMT Cable Off
21703	NMT Stimulus Off
21704	NMT Sensor Error
21705	NMT Low Battery
23001	Battery Low
23002	Battery Fault
23003	CMS Disconnected
23004	Otherbed Disconnected
23005	Recorder Out Of Paper
23006	Recorder Too Hot
23007	Recorder Error
23008	Recorder Does Not Exist
23009	Low Disk Space
23010	Disk Full
23011	System Clock Error
23012	Processor error
23101	ECG Abnormal Unplugged
23102	SPO2 Abnormal Unplugged
23103	SPO2L Abnormal Unplugged
23104	NIBP Abnormal Unplugged
23105	RESP Abnormal Unplugged
23106	CO2 Abnormal Unplugged
23107	ICG Abnormal Unplugged
23108	BIS Abnormal Unplugged
23109	EEG Abnormal Unplugged
23110	AA Abnormal Unplugged

23111	O2 Abnormal Unplugged
23112	N2O Abnormal Unplugged
23113	CO2 Abnormal Unplugged
23114	RM Abnormal Unplugged
23115	IBP1 Abnormal Unplugged
23116	IBP2 Abnormal Unplugged
23117	IBP3 Abnormal Unplugged
23118	IBP4 Abnormal Unplugged
23119	IBP5 Abnormal Unplugged
23120	IBP6 Abnormal Unplugged
23121	IBP7 Abnormal Unplugged
23122	IBP8 Abnormal Unplugged
23123	ART Abnormal Unplugged
23124	CVP Abnormal Unplugged
23125	PA Abnormal Unplugged
23126	RAP Abnormal Unplugged
23127	LAP Abnormal Unplugged
23128	ICP Abnormal Unplugged
23129	UAP Abnormal Unplugged
23130	BAP Abnormal Unplugged
23131	FAP Abnormal Unplugged
23132	UVP Abnormal Unplugged
23133	LV Abnormal Unplugged
23134	AO Abnormal Unplugged
23145	TA Abnormal Unplugged
23146	TB Abnormal Unplugged
23147	TC Abnormal Unplugged
23148	TD Abnormal Unplugged
23149	TE Abnormal Unplugged
23150	DM Abnormal Unplugged
23151	NMT Abnormal Unplugged
23152	P1 Abnormal Unplugged
23153	MPS Abnormal Unplugged
21101	T1 Self Checking Error
21102	T1 Sensor Off
21103	T1 Out High Limit
21104	T1 Out Low Limit
21111	T2 Self Checking Error
21112	T2 Sensor Off
21113	T2 Out High Limit
21114	T2 Out Low Limit
21121	T3 Self Checking Error
21122	T3 Sensor Off

21123	T3 Out High Limit
21124	T3 Out Low Limit
21131	T4 Self Checking Error
21132	T4 Sensor Off
21133	T4 Out High Limit
21134	T4 Out Low Limit
21141	T5 Self Checking Error
21142	T5 Sensor Off
21143	T5 Out High Limit
21144	T5 Out Low Limit
21151	T6 Self Checking Error
21152	T6 Sensor Off
21153	T6 Out High Limit
21154	T6 Out Low Limit
21161	T7 Self Checking Error
21162	T7 Sensor Off
21163	T7 Out High Limit
21164	T7 Out Low Limit
21171	T8 Self Checking Error
21172	T8 Sensor Off
21173	T8 Out High Limit
21174	T8 Out Low Limit
21181	T9 Self Checking Error
21182	T9 Sensor Off
21183	T9 Out High Limit
21184	T9 Out Low Limit
21191	T10 Self Checking Error
21192	T10 Sensor Off
21193	T10 Out High Limit
21194	T10 Out Low Limit
21201	Skin Temp Self Checking Error
21202	Skin Temp Sensor Off
21203	Skin Temp Out High Limit
21204	Skin Temp Out Low Limit
21211	Core Temp Self Checking Error
21212	Core Temp Sensor Off
21213	Core Temp Out High Limit
21214	Core Temp Out Low Limit
21221	Axil Temp Self Checking Error
21222	Axil Temp Sensor Off
21223	Axil Temp Out High Limit
21224	Axil Temp Out Low Limit
21231	Naso Temp Self Checking Error

21232	Naso Temp Sensor Off
21233	Naso Temp Out High Limit
21234	Naso Temp Out Low Limit
21241	Eso Temp Self Checking Error
21242	Eso Temp Sensor Off
21243	Eso Temp Out High Limit
21244	Eso Temp Out Low Limit
21251	Rect Temp Self Checking Error
21252	Rect Temp Sensor Off
21253	Rect Temp Out High Limit
21254	Rect Temp Out Low Limit
21261	Amb Temp Self Checking Error
21262	Amb Temp Sensor Off
21263	Amb Temp Out High Limit
21264	Amb Temp Out Low Limit
21271	Airw Temp Self Checking Error
21272	Airw Temp Sensor Off
21273	Airw Temp Out High Limit
21274	Airw Temp Out Low Limit
21281	Vesic Temp Self Checking Error
21282	Vesic Temp Sensor Off
21283	Vesic Temp Out High Limit
21284	Vesic Temp Out Low Limit
21291	Blood Temp Self Checking Error
21292	Blood Temp Sensor Off
21293	Blood Temp Out High Limit
21294	Blood Temp Out Low Limit
21301	Myo Temp Self Checking Error
21302	Myo Temp Sensor Off
21303	Myo Temp Out High Limit
21304	Myo Temp Out Low Limit
21311	Tymp Temp Self Checking Error
21312	Tymp Temp Sensor Off
21313	Tymp Temp Out High Limit
21314	Tymp Temp Out Low Limit
21321	Cereb Temp Self Checking Error
21322	Cereb Temp Sensor Off
21323	Cereb Temp Out High Limit
21324	Cereb Temp Out Low Limit
21501	RSO2 Communication Error
21502	RSO2_CH1 Out of Range
21503	RSO2_CH1 Short Path Symmetry
21504	RSO2_CH1 Differential symmetry

21505	RSO2_CH1 Light absorption out of range
21506	RSO2_CH1 Noise
21507	RSO2_CH1 Differential signal slope too large
21508	RSO2_CH1 Short path out of range
21509	RSO2_CH1 Long path too low
21510	RSO2_CH1 No Sensor
21511	RSO2_CH1 Unauthorized Sensor
21512	RSO2_CH1 Incompatible Sensor
21513	RSO2_CH1 Inadequate Power
21514	RSO2_CH1 Defective Sensor
21515	RSO2_CH1 No Cable Connected
21516	RSO2_CH1 Cable Incompatible
21517	RSO2_CH1 Defective Cable
21518	RSO2_CH1 Cable Will Expire Soon
21519	RSO2_CH1 Cable Life Expired
21520	RSO2_CH1 No Cable Connected
21521	RSO2_CH1 Defective Sensor
21522	RSO2_CH1 Sensor Life Near Expiration
21523	RSO2_CH1 Sensor Life Expired
21524	RSO2_CH1 No Adhesive Connected
21525	RSO2_CH1 Incompatible Adhesive
21526	RSO2_CH1 Defective Adhesive
21527	RSO2_CH1 Adhesive Life Near Expiration
21528	RSO2_CH1 Adhesive Life Expired
21529	RSO2_CH1 Sensor Initializing
21530	RSO2_CH1 Sensor Off
21531	RSO2_CH1 Pulse Search
21532	RSO2_CH1 Motion Interference
21533	RSO2_CH1 Low Perfusion Index
21534	RSO2_CH1 Check Sensor Connection
21535	RSO2_CH1 Demo Mode
21536	RSO2_CH1 No Sensor Site Selected
21552	RSO2_CH2 Out of Range
21553	RSO2_CH2 Short path symmetry
21554	RSO2_CH2 Differential symmetry
21555	RSO2_CH2 Light absorption out of range
21556	RSO2_CH2 Noise
21557	RSO2_CH2 Differential signal slope too large
21557	RSO2_CH2 Short path out of range
21558	RSO2_CH2 Long path too low
21560	RSO2_CH2 No Sensor
21561	RSO2_CH2 Unauthorized Sensor
21562	RSO2_CH2 Incompatible Sensor

21563	RSO2_CH2 Inadequate Power
21564	RSO2_CH2 Defective Sensor
21565	RSO2_CH2 No Cable Connected
21566	RSO2_CH2 Cable Incompatible
21567	RSO2_CH2 Defective Cable
21568	RSO2_CH2 Cable Will Expire Soon
21569	RSO2_CH2 Cable Life Expired
21570	RSO2_CH2 No Cable Connected
21571	RSO2_CH2 Defective Sensor
21572	RSO2_CH2 Sensor Life Near Expiration
21573	RSO2_CH2 Sensor Life Expired
21574	RSO2_CH2 No Adhesive Connected
21575	RSO2_CH2 Incompatible Adhesive
21576	RSO2_CH2 Defective Adhesive
21577	RSO2_CH2 Adhesive Life Near Expiration
21578	RSO2_CH2 Adhesive Life Expired
21579	RSO2_CH2 Sensor Initializing
21580	RSO2_CH2 Sensor Off
21581	RSO2_CH2 Pulse Search
21582	RSO2_CH2 Motion Interference
21583	RSO2_CH2 Low Perfusion Index
21584	RSO2_CH2 Check Sensor Connection
21585	RSO2_CH2 Demo Mode
21586	RSO2_CH2 No Sensor Site Selected

Appendix A.8 Module waveform ID

ID	内容
30001	ECG_1
30002	ECG_2
30003	ECG_3
30004	ECG_Avr
30005	ECG_Avl
30006	ECG_Avf
30007	ECG_V1
30008	ECG_V2
30009	ECG_V3
30010	ECG_V4
30011	ECG_V5
30012	ECG_V6

30013	ECG_Va
30014	ECG_Vb
30015	ECG_V
30021	SPO2
30022	SPO2L
30031	RESP
30032	ICG
30033	BIS
30041	EEG1
30042	EEG2
30043	EEG3
30044	EEG4
30051	CO
30052	CO2
30053	N2O
30054	O2
30061	AA
30062	HAL
30063	ENF
30064	ISO
30065	SEV
30066	DES
30071	RM_FLOW
30072	RM_PAW
30073	RM_VOL
30081	IBP_P1
30082	IBP_P2
30083	IBP_ART
30084	IBP_CVP
30085	IBP_PA
30086	IBP_RAP
30087	IBP_LAP
30088	IBP_ICP
30089	IBP_PAWP_NULL
30090	IBP_UAP
30091	IBP_BAP
30092	IBP_FAP
30093	IBP_UVP
30094	IBP_LV
30095	IBP_AO
30096	IBP_PAWP
30100	PR
30101	RR

Appendix A.9 Units ID

ID	Contents
11	mmHg
12	kPa
13	cmH2O
14	%
21	C
22	F
31	cm
32	inch
41	kg
42	lb

Note: Italics in the document are reserved items

Appendix A.10 Unit conversion

Original unit	Target unit	Transformation relation
mmHg（Millimetes of mercury）	kPa（Kilopascal）	1 mmHg = 0.13328 kPa
	cmH2O（Centimeters of water）	1 mmHg = 1.36 cmH2O
	%（Percentage）	1 mmHg = 100 / P P is the standard atmosphere
lb（Pound）	Kg（kilogram）	1 lb = 0.45359 Kg
in（Iinch）	cm（Centimeter）	1 in = 2.54 cm
ºC（Centigrade）	ºF（Fahrenheit）	F = 32 + C * 1.8 Where F is degrees Fahrenheit and C is degrees Centigrade
Drops/min（Drops/minutes	ml/h（Milliliter/hour）	Dropping speed (mL/h) = Liquid volume(mL) * Drops per minute(drops/min) ÷ 60

Note: Standard atmospheric pressure conversion P=1.013X10 ^ 5 (Pa)=760 (mmHg)